Few-Shot Target Detection in SAR Imagery via Intensive Metafeature Aggregation

Zheng Zhou, *Graduate Student Member, IEEE*, Zongjie Cao, *Member, IEEE*, Qin Chen, *Graduate Student Member, IEEE*, Kailing Tang, *Graduate Student Member, IEEE*, Yujian Li, *Graduate Student Member, IEEE*, Yiming Pi, *Senior Member, IEEE*, and Zongyong Cui, *Member, IEEE*

***Abstract*— Synthetic aperture radar (SAR) targets often exhibit characteristics, such as high mobility and strong concealment, resulting in scarce SAR data and the manifestation of few-shot data properties. These few-shot SAR targets are susceptible to interference from complex background information and mutual interference of target features, making it challenging to distinguish SAR targets from the background. In addition, there is confusion in features among different targets, leading to models being highly insensitive to few-shot SAR targets under complex distribution conditions in new tasks. Similarly, these few-shot SAR targets exhibit significant sample scarcity and sample variations, resulting in pronounced fluctuations in class centers and difficulty in determining sample distributions. This leads to challenges in accurately representing the potential representative features of few-shot SAR targets by the model. To address these issues, further enhancement of SAR target features is necessary to provide a robust foundation for the ultimate aggregation module. Therefore, based on the metalearning paradigm, we propose a method for few-shot target detection in SAR imagery via intensive metafeature aggregation (IMFA), aiming to reinforce SAR target features for improved representation. Specifically, first, we propose a novel hierarchical multihead cross attention (HMCA) to capture the global multiscale contextual information in different subspaces and analyze representative features between different targets to distinguish SAR targets from the background. Then, based on HMCA, we introduce a novel feature coupling module (FCM) to couple support features with cognitive information from the query image on the support branch. This is done to reduce the confusion and mutual interference of features between targets while enhancing the model's generalization ability on new tasks. Finally, on the support branch with query-aware information, we construct a Gaussian distribution to estimate the class distribution of few-shot SAR targets and replace traditional class prototypes. On this basis, we propose the feature information maximization module (FIMM) to the avoid feature information shift, greatly strengthening the expression of potential features. Through these steps, reinforced metafeatures can be obtained, enabling efficient aggregation. Experiments on the SRSDD-v1.0 and MSAR-1.0 datasets demonstrate that our method has consistently outperformed state-of-the-art approaches in all configurations, achieving state-of-the-art performance.**

***Index Terms*— Few-shot target detection, intensive metafeature aggregation (IMFA), metalearning, metric learning, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR), as an active remote sensing system working at microwave frequencies, can obtain high-resolution 2-D images [1], [2], and it also carries substantial significance in various fields, such as national defense, environmental protection, and Earth sciences [3], [4]. Meanwhile, studies dedicated to the interpretation of high-resolution SAR images have become increasingly important, with SAR image target detection as a critical research subject. The study of traditional SAR target detection is mainly centered on the constant false alarm rate (CFAR) algorithm [5], [6], [7]. The algorithm detected the target pixel by comparing each pixel's grayscale with the specific detection thresholds. The settings of thresholds are determined by the statistical properties of background clutter, to enhance the target detection accuracy while maintaining a CFAR. However, when the image scenarios become complex, CFAR methods often fail to achieve a better detection performance.

In recent years, the development of deep learning (DL), [8], [9], [10], [11], has pushed target detection to a higher new level, with the emergence of many excellent target detection algorithms [12], [13], [14], [15], [16]. Considering the unique characteristics of SAR images, some scholars designed a target detection algorithm exclusively for SAR images [17], [18], [19], [20], enormously boosting the accuracy and efficiency of SAR target detection. However, these detection algorithms apply data-driven methods, heavily relying on the data scale and statistics distribution characteristics of SAR target samples. They also present a sudden fall in the detection performance when dealing with long-tail objects (typically with a tiny number of annotated samples) [21]. Under such information-constrained circumstances, how to achieve SAR target detection is a typical few-shot target detection problem.

Manuscript received 12 December 2023; revised 17 April 2024 and 18 May 2024; accepted 19 May 2024. Date of publication 27 May 2024; date of current version 10 June 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271116 and Grant 61971101 and in part by the Cooperative Construction Project in Yibin under Grant DZKJDX2021020001. *(Corresponding author: Zongyong Cui.)*

Zheng Zhou, Qin Chen, Kailing Tang, Yujian Li, Yiming Pi, and Zongyong Cui are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zhengzhou_uestc@163.com; qinchen_uestc@163.com; 907015270@qq.com; liyujianjiangshi@163.com; ympi@uestc.edu.cn; zycui@uestc.edu.cn).

Zongjie Cao is with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the Intelligent Terminal Key Laboratory of Sichuan Province, Yibin, Sichuan 644005, China (e-mail: zjcao@uestc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3405637

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

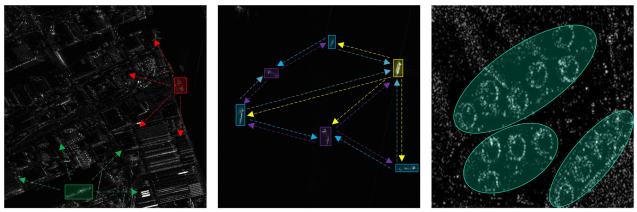
The core is to neutralize the contradiction between the limited supplement of SAR target sample data and the data requirement of the detection model, so as to reduce the heavy reliance of the data-driven target detection algorithm on the scale of SAR sample data and its distribution characteristics.

To address the problems including the sample scarcity faced by current methods, few-shot learning has been proposed and widely applied in various tasks in DL [22], [23]. Some common target detection algorithms are combined with few-shot learning approaches contributing to acceptable detection performance [24], [25], [26], [27]. For instance, Fan et al. [24] proposed an attention-based regional proposal network (RPN) module and a multirelation detector, which enhance the localization accuracy and classifying ability. Sun et al. [22] advanced a simple and effective method of contrastive-aware object proposal encodings. In addition, Qiao et al. [25] introduced the gradient decoupling layers of multistage decoupling and a prototypical calibration block for multitask decoupling aiming to enhance the performance. Recently, Li et al. [26] proposed a novel architecture called AirDet without fine-tuning, which can learn and support the class-agnostic relationships throughout all modules, including cross-scale object proposal networks, aggregation modules, and localization networks. Meanwhile, Zhang et al. [27] designed a metrics-based discriminative loss to learn more discriminative classifiers during the few-shot fine-tuning stage.

However, the abovementioned few-shot target detection algorithms are not designed particularly for SAR datasets, yet do not consider whether they could effectively extract feature information of targets from novel classes and how to intensify the extracted feature information when faced with novel class targets. Thus, applying these methods directly to SAR datasets may cause problems such as a significant drop in their detection performance.

Though DL has a powerful data representation ability, training deep neural network models that can sufficiently extract features from SAR datasets with limited annotated samples could still be quite tough. There are four main causes.

1) The collection of SAR data is more challenging than that of natural images. SAR target labeling is costly and time-consuming, leading to a scarcity of SAR samples.
2) SAR images, with lower resolution than optical images and susceptibility to clutter interference, face challenges in extracting meaningful features due to the limited SAR samples and the variability in target categories compared with optical images.
3) SAR targets, especially high-value strategic targets, are often equipped with high mobility and strong concealment capabilities. This tremendously increases the difficulty of acquiring SAR images of such targets, leading to a severe shortage of such targets' samples.
4) Battlefield situations often exhibit considerable uncertainty, and the frequency of different SAR targets varies significantly across diverse scenarios. Due to limitations in SAR sensors, obtaining an equal amount of data for each target sample is challenging, leading to imbalances in sample numbers among different classes, with some classes experiencing severe scarcity of samples.



(a) (b) (c)

Fig. 1. Practical challenges in few-shot SAR target detection are evident. (a) SAR target feature is susceptible to strong interference from complex background features, challenging the model's ability to distinguish between target and background. (b) Interclass similarity and intraclass variation among different SAR target classes, causing confusion in features and inaccurate model distinctions between base and novel classes. (c) Diverse and complex distributions of SAR targets, leading to the model's lack of accurate knowledge regarding sample distribution and few-shot target features.

Consequently, these four factors contribute to a sudden fall in the number of SAR target samples of some classes, and the SAR target detection performance on these deficient samples is correspondingly worse, even leading to a sharp decrease in overall detection performance simultaneously.

There are currently numerous few-shot target classification algorithms [28], [29], [30], [31], [32], [33], but only few SAR specialized few-shot target detection algorithms are existed [34], [35], [36]. For example, Wang et al. [34] realize the performance enhancement primarily through applying metric learning on classifiers during the fine-tuning stage, and Chen et al. [35] use attention mechanisms and dynamic relationship learning paradigms to improve feature representation. In addition, Zhou et al. [36] advance a transformer-based aggregator to enhance the correlation between different categories in the support set. However, these methods did not take into account the severe scarcity and characteristics of SAR samples, they merely made preliminary adjustments to optical domain methods. Moreover, the performance of few-shot SAR target detection can be influenced by complex background interference. Strong confusion may also arise among different SAR target classes due to feature similarity. In addition, in complex scenarios, the diverse distribution characteristics of SAR targets may hinder the model's adaptation to new tasks, resulting in decreased performance for novel class targets, as shown in Fig. 1.

Therefore, following the metalearning paradigm, we propose a method for few-shot target detection in SAR imagery via intensive metafeature aggregation (IMFA). The goal is to reinforce SAR target features for better representation. This method efficiently utilizes SAR target features, delivering outstanding performance in scenarios with extremely limited SAR samples, especially for few-shot novel-class targets. This article's main contributions are sixfold.

1) We access the new paradigm of the few-shot SAR target detection by intensifying SAR target features. We propose a few-shot target detector based on the metalearning paradigm, specifically designed for SAR scenarios.
2) We present a novel hierarchical multihead cross attention module (HMCA) that fully considers issues in SAR images including insufficiency of contextual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

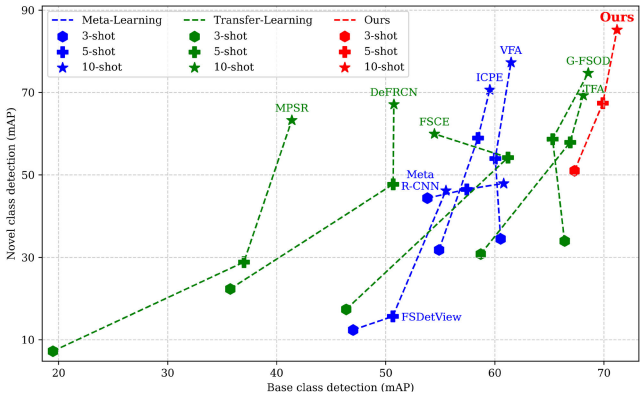


Fig. 2. Based on the SRSDD-v1.0 dataset, we compared the base-novel detection performance of IMFA with other state-of-the-art FSOD methods in split 1 setting with $K = \{3, 5, 10\}$ shots. As shown in the figures, our detection performance outperforms other algorithms under the conditions of 3, 5, and 10 shots.

relationships between targets and background as well as scarcity of feature information. The method realizes the precise extraction of representative features for SAR images via capturing global multiscale contextual information and the relationship between different target features in various subspaces.

3) We propose a novel feature coupling module (FCM) that builds a feature coupling bridge between query features and support features. Coupling feature of query features onto support feature branches provides additional contextual information, assisting the detection model in better distinguishing targets from distinctive categories and mitigating the intraclass variations. Meanwhile, it facilitates the detection model to better capture specific features of SAR targets in query images and rapidly adapt to novel-class targets via shared information in support sets, thereby enhancing the detection performance.
4) We propose a novel feature information maximization module (FIMM). Initially, the traditional SAR target class prototypes are transformed into class distributions with Gaussian forms through encoders, reducing fluctuations situated in the class center of the SAR target under few-shot conditions. Subsequently, we establish the feature information maximization bridge between support features and latent features as a constraint function, eliminating the imbalance between support features and latent feature information and avoiding feature information biases. This realizes maximization aggregation of latent features.
5) Our proposed IMFA algorithm achieves state-of-the-art few-shot target detection performance on the publicly available SRSDD-v1.0 and MSAR-1.0 datasets, attaining the most competitive performance for both base-class and novel-class targets (as shown in Fig. 2).

The subsequent sections are summarized as follows. Section II reviews the pertinent literature on SAR target detection and few-shot target detection. Section III offers preliminary knowledge and motivation for our proposed method. Section IV provides a detailed description of the innovative modules. Section V details the experiments on two public SAR datasets using our method and presents a summary and analysis. Finally, Section VI offers the conclusive remarks on our proposed method.

II. Related Works

A. SAR Target Detection

In SAR image interpretation, the target detection is a crucial and challenging task. Two primary types of detectors based on model structures are convolutional neural network (CNN)-based and transformer-based algorithms. Researchers have enhanced and extended these detectors to suit SAR images, significantly advancing SAR target detection in this domain.

SAR target detection algorithms [18], [37], [38] based on CNN extract features via convolutional operations within local receptive fields, effectively capturing the texture and edge information of targets in SAR images. For example, Deng et al. [37] designed a condensed backbone network composed of multiple dense blocks, achieving target localization by predicting multiscale ship candidate boxes from various intermediate layers. Meanwhile, Zhang et al. [18] proposed a novel balanced learning network to address the unnoticed imbalances that remain in SAR ship detectors. Also, Cui et al. [38] proposed a framework based on the threshold neural network to separate targets by predicting the optimal detection threshold for sliding windows.

On the other hand, SAR target detection algorithms based on transformer [39], [40] globally model the entire image using self-attention mechanisms, effectively capturing the long-range dependencies and correlations between targets in SAR images. For instance, Zhou et al. [39] proposed a SAR ship detection framework based on the pyramid vision transformer paradigm, overcoming limitations in the receptive field to address issues such as data scarcity. In addition, Li et al. [40] proposed an enhancement Swin transformer detection network to improve the utilization and localization capabilities of ship information in SAR images.

As mentioned earlier, numerous SAR target detectors perform exceptionally well with abundant annotated samples. However, obtaining large batches of SAR data with annotations, especially for high-value targets, is challenging and time-consuming. These detectors may exhibit suboptimal performance under such circumstances.

B. Few-Shot Object Detection

In response to challenges such as insufficient annotated samples, some researchers have proposed few-shot object detectors to address such issues. From the perspective of working principles, these detectors can be categorized into methods based on data augmentation, metalearning, transfer learning, and metric learning.

The few-shot object detection (FSOD) problem [41], [42] is primarily addressed through metalearning and transfer learning. These paradigms are chosen due to their ability to transfer prior knowledge to novel class targets, mitigating the challenge of limited samples for such targets. For example,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

Zhang et al. [43] employed a metalearning strategy to capture interclass correlations from two perspectives: low-quality region proposals and interclass relevance, reducing misclassifications and enhancing generalization. Xiao et al. [44] extracted class-representative feature information from data from different modes to instruct network predictions to enhance FSOD performance. However, Lu et al. [45] advanced the dynamic aggregation module from the perspective of feature prototypes to dynamically adjust intraimage and interimage weights to emphasize information in query images. Han et al. [46] proposed a class-agnostic aggregation method and a class category-related aggregation method in view of feature aggregation, reducing interclass confusion while enhancing feature expressions. In addition, Wang et al. [47] proposed a two-stage fine-tuning approach (TFA), which involves initializing only the weights of the final layer of the network, and then fine-tunes the bounding box classification and regression network on class-balanced few-shot base-class and novel-class training sets. Moreover, Sun et al. [22] used few-shot contrastive proposal encoding (FSCE) to enhance instance-level intraclass compactness and interclass distinctiveness, alleviating misclassification issues. Yang et al. [42] proposed an effective pretraining transfer framework baseline and introduced an initializer called knowledge inheritance to initialize new weights of the anchor box classifier, efficaciously facilitating knowledge transfer and boosting self-adaptive speed.

However, algorithms of the above-mentioned different paradigms do not apply to SAR targets. The challenges in SAR scenarios arise from the complex background information interference and severely poor target feature information in SAR scenarios, terrible feature information interference between and within SAR target classes, and complicated sample distribution characteristics. Our approach belongs to the paradigm of metalearning, and it is pioneering in utilizing SAR target characteristics and feature information to enable few-shot target detection in the SAR domain.

III. Preliminary Knowledge

A. Problem Definition

Similar to the previous FSOD methods [22], [47], [48], we strictly adhere to the standard FSOD problem settings. Initially, we need to confirm a complete dataset called $D = \{(x_i, y_i), y_i \in C\}_{i=1}^{n}$, where x represents the input image and y represents the corresponding annotations, and then randomly split this dataset D into a base class dataset $D_{\text{base}} = \{(x_i^{\text{base}}, y_i^{\text{base}}), y_i^{\text{base}} \in C_{\text{base}}\}_{i=1}^{n_{\text{base}}}$ and a novel class dataset $D_{\text{novel}} = \{(x_i^{\text{novel}}, y_i^{\text{novel}}), y_i^{\text{novel}} \in C_{\text{novel}}\}_{i=1}^{n_{\text{novel}}}$ according to the category. Each base class in D_{base} contains sufficient samples, while each novel class in D_{novel} contains very few $K = \{1, 2, 3, \ldots\}$ samples. It is particularly stressed that the base class dataset D_{base} must not overlap with the novel class dataset D_{novel}, i.e., $C_{\text{base}} \cup C_{\text{novel}} = C$ and $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. In addition, our few-shot target detection task follows a two-stage training paradigm, including the base training stage and the few-shot fine-tuning stage. The base training stage does not contain novel-class samples, while the fine-tuning stage equivalently contains samples from base classes and novel classes. Meanwhile, we adapt to FSOD tasks according to the metalearning paradigm [49], ensuring that the model rapidly adapts or generalizes to new tasks.

B. Motivations and Baseline Method

As shown in Fig. 1, we start from the characteristics of SAR targets to explore the practical challenges faced by SAR targets in complex scenes. We summarize that the challenges in the following three specific domains are the reasons that make the few-shot SAR target detection task more challenging.

1) *Complex Background Information Interference and Severely Poor Target Feature Information:* The correlation between the complicated background information and specific target information is strong. Since there is fierce clutter around targets, such complex contextual information critically impacts the accuracy representation of the detection model and confuses the feature information between targets and backgrounds. Therefore, capturing the representative features of targets requires deep and efficient analysis of the contextual information of SAR images, to achieve the deep extraction of presentative features of SAR targets.
2) *Severe Feature Information Interference Between and Within SAR Target Classes:* Due to the potential feature confusion among different target classes in the feature space, as well as significant feature differences among samples within the same class, the interference of these features hinders detection models from better understanding target characteristics and the relationships between samples, greatly impacting the generalization of knowledge from base classes to novel class targets. Thus, reducing the interference of feature information while enhancing its transmission requires building feature information bridges between different classes, assisting the detection model to better capture specific features of candidate targets.
3) *Complicated Sample Distribution Characteristics:* Because of combined influences of diversity, geographical and temporal alterations, occlusion and reflection differences, target density, and class imbalance, the distribution characteristics of SAR samples become highly complex, and the target distribution of different regions is not uniform as well. Therefore, capturing more effectively latent features of SAR target samples requires thoroughly analyzing its complex distribution form and exploring its additional feature distribution information. This realizes the maximized use of SAR target feature information, which is hugely favorable for few-shot target detection with scarce samples.

Countering the aforementioned challenges in specific fields, it is urgent to unlock the few-shot target detection in the SAR field and address its unique and practical problems. Considering the capability of rapid adaptations of the metalearning paradigm when facing new tasks, we adopt the latest FSOD algorithm named variational feature aggregation (VFA) [46]

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

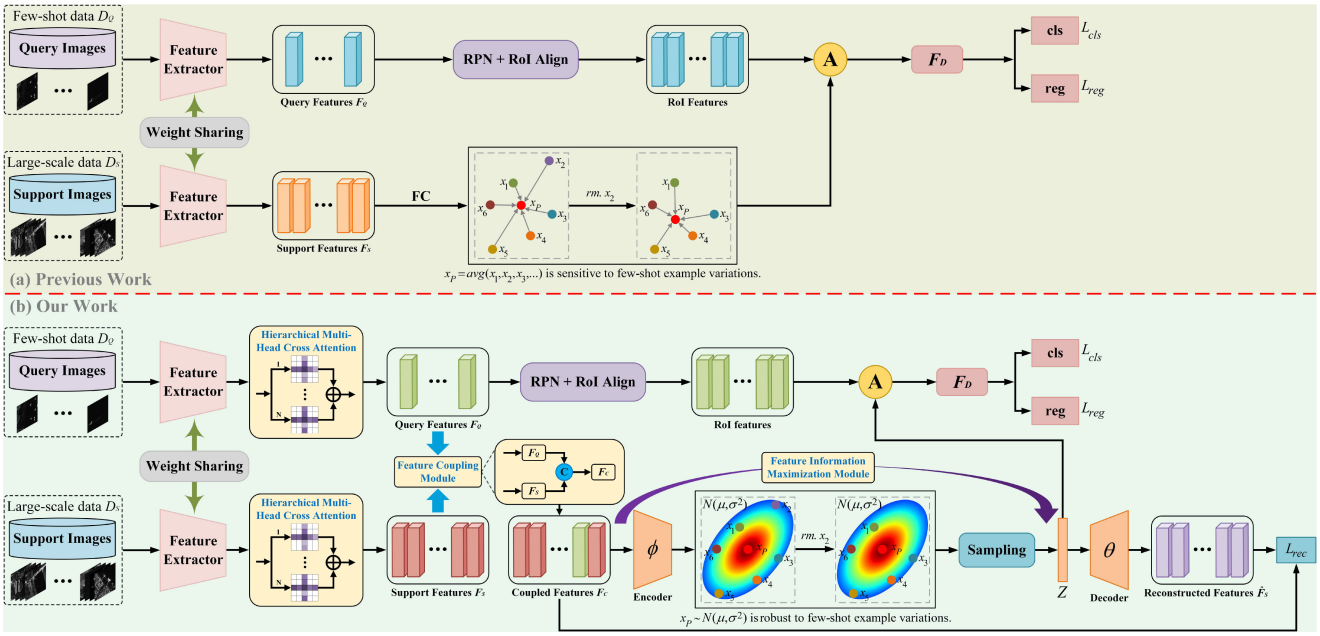


Fig. 3. Comparison between our proposed IMFA and conventional methods. (a) Conventional architecture of previous methods and (b) overall architecture of our proposed method. In prior approaches, the core involved obtaining class prototypes by averaging all features and subsequently performing feature aggregation. In our work, we first construct an HMCA module with weight sharing to extract representative features of SAR targets. Subsequently, we couple the feature from the query branch with the support branch to enhance the recognition of novel-class targets. Finally, we introduce the Gaussian distributions on the support features to transform class prototypes into class distributions. Simultaneously, we establish feature information bridges to maximize latent feature expression. Through the above steps, intensive features of SAR targets can be captured, ultimately achieving efficient feature aggregation.

as the baseline of this article. However, when confronted with the unique and practical problems mentioned above, directly employing the VFA algorithm in SAR scenarios will exhibit poor performance. Therefore, we have designed an efficient metalearning architecture for few-shot SAR target detection under incomplete data conditions.

IV. Proposed Method

In this section, we first provide a detailed overview of the overall architecture of the proposed method, comparing it with conventional approaches used in the past. Next, we offer detailed descriptions of each innovative module introduced in our proposal. Finally, we present the overall loss function.

A. Overall Architecture of IMFA

As illustrated in Fig. 3, we present a detailed architecture of our proposed method, while also comparing it with the conventional architectures of previous methods.

The conventional methods in the past, as shown in Fig. 3(a), neither took into account the contextual feature information and the relevance of the background for all targets nor consider the feature information of the query branch. Moreover, this paradigm often suffers from severe intraclass centroid fluctuation issues, particularly in scenarios with extremely limited samples or substantial sample variations, resulting in poor aggregation outcomes. Furthermore, SAR target feature are frequently influenced by background clutter, interclass similarities, intraclass variances, and distribution characteristics. In the face of scenarios with complex backgrounds, few samples, and targets with fine-grained features, this approach struggles to effectively extract representative features of the targets and accurately detect novel-class targets.

The proposed IMFA [as shown in Fig. 3(b)] focuses on enhancing the representative features of SAR targets by addressing three aspects: extraction, coupling, and maximization of metafeatures. This results in the efficient aggregation of SAR target features. When encountering novel-class targets, accurate decisions can be made based on representative features.

In particular, we introduce an HMCA mechanism into the original backbone network. This construction aims to establish a feature extractor capable of capturing contextual information and correlations among different SAR targets, to reduce the interference of complex SAR background information on target-specific features and obtain more representative features. Furthermore, we build feature information bridges between the query and support branches to couple the feature information of the query image with the support features, enhancing the generalization of base-class knowledge and awareness of novel-class targets.

On the other hand, inspired by variational feature learning [50], we employ a carefully constructed encoder to generate the latent variable Z in the form of a multivariate Gaussian distribution from the support features F_S, avoiding the issue of class center fluctuation caused by sample scarcity. Considering the significant dimension difference between support features and latent variables, along with the improper inference of the evidence lower bound (ELBO), which leads to the model sacrificing essential information of latent variables and severely hindering feature aggregation's input information. Drawing inspiration from mutual information learning [51],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

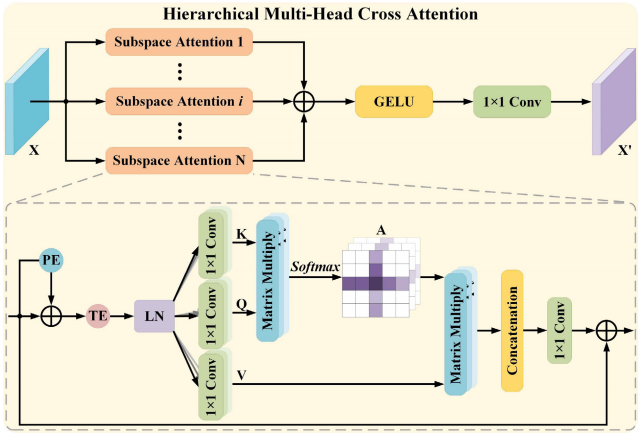


Fig. 4. Schematic of HMCA. Initially, the input features are effectively partitioned into different subspaces by channel. Subsequently, multihead cross attention operations are executed separately in these different subspaces. Finally, the extracted SAR target feature is efficiently fused to obtain SAR target features that are more representative.

[52], we establish feature information bridges between input support features and generated latent variables to eliminate the model's information bias problem, thus obtaining more valuable latent features and achieving more robust feature aggregation.

Consequently, compared with previous approaches, we have accomplished the following key functionalities.

1) From the perspective of the SAR target feature, we have efficiently achieved more systematic feature aggregation using extraction, coupling, and maximization of utility.
2) The hierarchical encoding of multihead cross attention enables the selective extraction of global contextual features from SAR images, facilitating the extraction of representative features for SAR targets.
3) We have enhanced the awareness of feature for query features by coupling the feature of query images with support branches, thereby improving the accurate classification of query targets and novel-class targets.
4) We have introduced a novel conceptual transformation, converting conventional class prototypes into Gaussian distributions to stabilize class centers.
5) Efficiently expressing latent features through the adoption of a feature information maximization constraint function.

B. Hierarchical Multihead Cross Attention Module

Due to the strong correlation between the complex background information in SAR images and the distinctive SAR target features, contextual information of the target significantly affects the accurate representation of the detection model. Inspired by works such as [53] and [54], we capture the multiscale global contextual information from various subspaces and analyze the relationships between different target representative features to distinguish between targets and backgrounds. To address this, we introduce the novel HMCA module. HMCA performs various tasks in different channels' subspaces, primarily capturing diverse data space relationships and collecting multiscale contextual information both horizontally and vertically to enhance the pixel-level expressiveness.

As illustrated in Fig. 4, given a local feature map $X \in \mathbb{R}^{C\times W\times H}$, this module first uniformly divides it into $C_i = \{C_1, C_2, \ldots, C_n\}_{i=1}^{n}$ on channel C by even and regular intervals, corresponding to local features $X_i = \{X_1, X_2, \ldots, X_n\}_{i=1}^{n}$. Each C_i exists independently and performs subattention tasks independently. For each subattention module, this module applies position encoding (PE) and task encoding (TE) to X_i, followed by a normalization layer (LN). Then, three convolution layers with 1×1 filters are applied to the output, generating three feature maps K_m, Q_m, and V_m. They can be defined as follows:

$$K_m = \mathrm{Conv}_K^{1\times1}\left(\mathrm{LN}\left(\mathrm{TE}\left(\mathrm{PE}\left(X_i^K\right)\right)\right)\cdot W_m^K\right) \tag{1}$$

$$Q_m = \mathrm{Conv}_Q^{1\times1}\left(\mathrm{LN}\left(\mathrm{TE}\left(\mathrm{PE}\left(X_i^Q\right)\right)\right)\cdot W_m^Q\right) \tag{2}$$

$$V_m = \mathrm{Conv}_V^{1\times1}\left(\mathrm{LN}\left(\mathrm{TE}\left(\mathrm{PE}\left(X_i^V\right)\right)\right)\cdot W_m^V\right) \tag{3}$$

where $\{K_m, Q_m\} \in \mathbb{R}^{C_i'\times W\times H}$, $V_m \in \mathbb{R}^{C\times W\times H}$; W_m^K, W_m^Q, and W_m^V represent the corresponding weights, and m denotes the number of multiple heads. Furthermore, C' represents the number of channels, which is smaller than C and is used for dimensionality reduction.

After obtaining K_m and Q_m, we further derive the attention map $A_m \in \mathbb{R}^{(H+W-1)\times(W\times H)}$ through matrix multiplication and a softmax operation [53]. At each position p along the spatial dimension of Q_m, we obtain a vector $Q_m^p \in \mathbb{R}^{C_i'}$. Similarly, we extract feature vectors from K_m that correspond to the same row or column as position p to obtain the set $\Omega_m^p \in \mathbb{R}^{(H+W-1)\times C_i'}$. $\Omega_{j,m}^p \in \mathbb{R}^{C_i'}$ represents the jth element of Ω_m^p. The definition of matrix multiplication is as follows:

$$d_{j,m}^p = Q_m^p \cdot {\Omega_{j,m}^p}^{\top} \tag{4}$$

where $d_{j,m}^p \in D_m$ represents the degree of correlation between features Q_m^p and $\Omega_{j,m}^p$, $j = [1, 2, \ldots, H + W - 1]$, and $D_m \in \mathbb{R}^{(H+W-1)\times(W\times H)}$, $\top$ is the transpose operation. Then, a softmax operation is applied along the channel dimension of D_m to calculate the attention map $A_{j,m}^p$

$$A_{j,m}^p = \frac{\exp\left(d_{j,m}^p\right)}{\sum_{j=0}^{H+W-1}\exp\left(d_{j,m}^p\right)}. \tag{5}$$

After obtaining the attention map $A_{j,m}^p$, we perform matrix multiplication between $A_{j,m}^p$ and V_m to obtain a feature vector $H_m \in \mathbb{R}^{C\times W\times H}$ that captures the global contextual information of the SAR image. This can be defined as follows. After obtaining the attention map $A_{j,m}^p$, we similarly take $V_m^p \in \mathbb{R}^C$ and the feature vector set $\Psi_m^p \in \mathbb{R}^{(H+W-1)\times C}$ corresponding to each position p in the spatial dimension of V_m. $\Psi_{j,m}^p$ is the set of feature vectors extracted from V_m that corresponds to the same row or column as position p. The attention map $A_{j,m}^p$ can selectively aggregate global contextual information, as shown in Fig. 5. Then, we perform matrix multiplication between $A_{j,m}^p$ and $\Psi_{j,m}^p$ to obtain a feature vector $H_{j,m}^p \in \mathbb{R}^{C\times W\times H}$ that captures the global contextual information of the SAR image. This can be defined as follows:

$$H_m^p = \sum_{j=0}^{H+W-1} A_{j,m}^p \cdot \Psi_{j,m}^p. \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

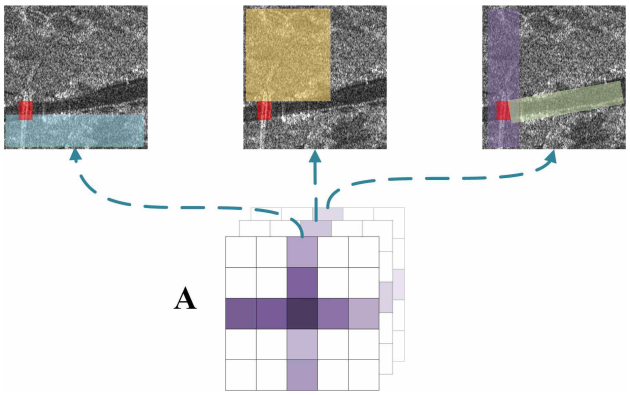


Fig. 5. Principle of the attention map A. A can perform the information extraction on the contextual multiscale information of specific targets in the global SAR image. The red region is the SAR bridge target and the other regions are the background. "A" can repeatedly extract the features of bridge targets in different subspaces, enhancing the model's attention to bridge targets while diminishing its focus on the background area, thereby distinguishing the targets from the background.

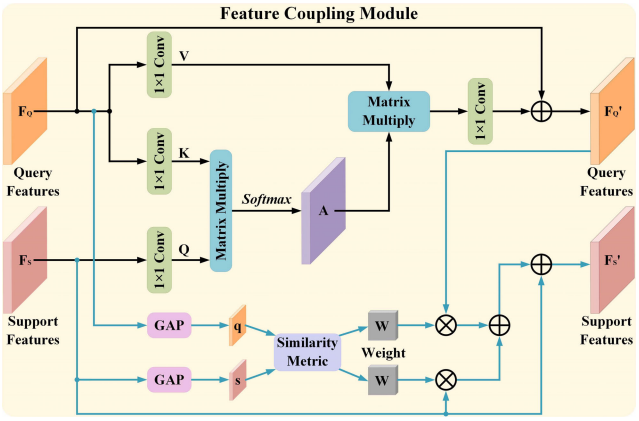


Fig. 6. Architecture of FCM. First, FCM generates a reorganized query feature F'_Q that encompasses features from both the query and support images. Subsequently, FCM predicts the global similarity weights W for support feature regions that are similar to the query feature. Finally, guided by W, F'_Q is dynamically coupled with F_S to form the new support feature F'_S.

According to our model design, it is necessary to merge the outputs from different heads, followed by a 1×1 convolution and residual connections within this subspace [8]. We define the output as Z_i, as defined specifically as follows:

$$Z_i = \mathrm{Conv}_h^{1\times1}\left(\sum_{s=1}^{m} H_m^p\right) + X_i. \tag{7}$$

Furthermore, the outputs Z_i from different subspaces are merged, and ultimately, the final feature X' is obtained through GELU and a 1×1 convolution, defined as follows:

$$X' = \mathrm{Conv}_x^{1\times1}\left(\mathrm{GELU}\left(\sum_{i=1}^{n} Z_i\right)\right). \tag{8}$$

Through these steps, we can incorporate contextual information from SAR images into the output X', enhancing pixel representations and the expression of specific SAR target features. Therefore, this module has the capability to selectively aggregate contextual information in the feature space based on different subspaces, thereby enhancing the robustness of attention maps for extracting representative SAR target features, as shown in Fig. 5.

C. Feature Coupling Module

Different SAR targets exhibit severe information interference, stemming from the interclass similarity and intraclass variability of SAR targets. This results in the model's insensitivity to the distinctive features of specific SAR targets. This limitation arises due to the inadequacy of representative features extracted from the support features and insufficient awareness of the query image, rendering the model incapable of adapting to various query images and diminishing the generalization from base-class knowledge to novel-class knowledge. Drawing inspiration from various works such as [55], we have devised a novel FCM to establish a bridge for feature information. FCM enhances the model's sensitivity to the representative information within the support features in SAR images and rectifies issues related to the model's insufficient awareness of query images.

As illustrated in Fig. 6, FCM mainly consists of an information coupling mechanism and a global similarity weight mechanism. Given the input features, the query feature F_Q and the support feature F_S, where F_Q undergoes convolutional layers with two 1×1 filters to generate the feature maps $V \in \mathbb{R}^{C\times W\times H}$ and $K \in \mathbb{R}^{C\times W\times H}$, and F_S passes through a convolutional layer with a 1×1 filter to produce the feature map $Q \in \mathbb{R}^{C\times W\times H}$, which can be defined as follows:

$$V = \mathrm{Conv}_V^{1\times1}(F_Q),\ \ K = \mathrm{Conv}_K^{1\times1}(F_Q),\ Q = \mathrm{Conv}_Q^{1\times1}(F_S). \tag{9}$$

On one hand, upon obtaining V, K, and Q, we first calculate the affinity between K and Q [53]. This results in the determination of the relevance between the support and query features. Subsequently, we employ a softmax operation to derive an attention matrix A with affinity weights. For each position p in the spatial dimensions of K and V, we obtain a vector $\{K^p, V^p\} \in \mathbb{R}^{C'}$, where $V^p \in \mathbb{R}^{C'}$ represents the ith element of K, and $V_i^p \in \mathbb{R}^{C'}$ represents the ith element of V. Similarly, we extract a feature vector Q^p from Q, where $Q_j^p \in \mathbb{R}^{C'}$ denotes the jth element of Q. The computation of A is as follows:

$$A_{i,j}^p = \frac{\exp\left(K_i^{p\top} \cdot Q_j^p\right)}{\sum_{i,j=0}^{H+W-1} \exp\left(K_i^{p\top} \cdot Q_j^p\right)} \tag{10}$$

where the affinity's of K and Q are computed by the compatibility function, while A can reflect the degree of association between different elements in the input SAR target features, and also provide global context-aware information for the SAR image. $i, j = [1, 2, \ldots, H + W - 1]$, and $\top$ is the transpose operation.

Furthermore, by multiplying A with V, we obtain a reconfigured feature vector H that concurrently encompasses feature information from both the query and support images. Finally, H undergoes a 1×1 convolutional layer and residual connections [8] to form a new customized query feature F'_Q. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

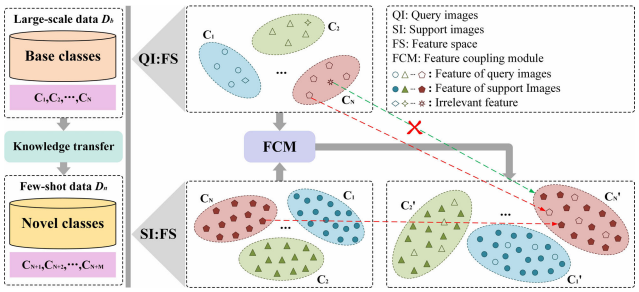


Fig. 7. Working principle of FCM. Both the base-class dataset D_b and the novel-class dataset D_n participate in the entire FCM process of the few-shot SAR target detection algorithm. Within the feature space, guided by FCM, representative features from the query image can be coupled with the support features. For instance, C'_N contains similarity-based feature between the query image C_N and the supported image C_N while excluding irrelevant features.

computation of F'_Q is as follows:

$$F'_Q = \mathrm{Conv}_q^{1\times 1}\left(\sum_{i,j=0}^{H+W-1} V_i^p \cdot A_{i,j}^p\right) + F_Q \tag{11}$$

where the reconstituted F'_Q already contains feature information that can be adapted to support features.

On the other hand, we compute a global similarity weight mechanism for SAR targets, considering F_Q and F_S. This mechanism predicts support feature regions highly consistent with the query feature as our global similarity weight, denoted by W, guiding the perfect integration of the reconstructed query feature F'_Q with the support feature, resulting in SAR support features F'_S enriched with query-aware information. We employ a global representation for both F_Q and F_S, utilizing the global similarity weight mechanism to analyze regions in F_S that exhibit high similarity to F_Q. The computation of the global similarity weight mechanism is as follows:

$$W\left(F_Q, F_S\right) = \frac{\mathrm{GAP}\left(F_Q\right) \cdot \mathrm{GAP}(F_S)}{\|\mathrm{GAP}\left(F_Q\right)\| \cdot \|\mathrm{GAP}(F_S)\|} \tag{12}$$

where GAP represents the global average pooling, and W is computed through cosine similarity. W will be used as feature guidance to activate the similarity regions of F'_Q and F_S for efficient feature information coupling. The definition of the restructured support feature F'_S is as follows:

$$F'_S = W \odot F'_Q + W \odot F_S + F_S \tag{13}$$

where $\odot$ represents the Hadamard product and F'_S maintains a residual connection [8] throughout the entire feature information coupling process.

To better understand the working principle of FCM, as shown in Fig. 7. FCM emphasizes positions with high similarity, assigning them greater weights, thereby directing the model's attention to locations in F'_S that are more similar to the query feature F_Q. Simultaneously, FCM attenuates dissimilar positions by assigning them smaller weights, reducing the influence of locations in F'_S that are dissimilar to the query feature F_Q. This allows the model to dynamically adjust the coupling of feature information to achieve efficient generalization and detection performance.

Compared with the initial F_S, F'_S is generated through deep interaction and effective coupling between F_Q and F_S, endowing F'_S with representative features from F_Q and F_S. This significantly enhances the recognition of SAR query features in new tasks and features of novel-class SAR targets, thus providing substantial assistance for the subsequent acquisition of latent features from SAR targets.

D. Feature Information Maximization Module

Previous approaches [44], [45] have achieved FSOD based on metalearning. However, these works employ fully connected layers to average these features to obtain class prototypes, without considering the distribution and transmission efficiency of feature information. This overlooks whether feature information are maximally utilized.

Motivated by variational feature learning [50], we explore the utilization of sample distribution and feature information of SAR targets from an unsupervised learning perspective. We introduce a novel encoder to learn meaningful latent features that depend on the input support features. Therefore, maximizing the utilization of meaningful support feature information becomes crucial, reducing the risk of information bias. Consequently, an FIMM based on the Gaussian distribution is introduced as an information bridge. Simultaneously, FIMM serves as a constraint function to alleviate the information imbalance issue between support and latent features, thereby enhancing the efficient expression and aggregation of latent features.

First, we assume that the input features are denoted by $F_S = \{s_1, s_2, \ldots, s_n\}_{i=1}^n$, where each data sample s_i is a randomly generated and mutually independent distributed variable. The reconstructed features are denoted by $\hat{F}'_S$, and the latent variables generated during the encoding process are denoted by $Z = \{z_i = \phi(s_i)\}_{i=1}^n$, ϕ is the encoder. The true data distribution is denoted by $p_D(s)$. Here, F_S is generated from the original complex conditional distribution $p_\theta(s|z)$, which is parameterized by a neural network, and θ is the decoder. Second, we also assume that the prior distribution of z, denoted by $p(z) = N(0, I)$, is a centered isotropic multivariate Gaussian distribution, and the posterior distribution denoted by $p_\theta(z|s)$ is a multivariate Gaussian distribution with diagonal covariance. The parameters $\mu = \{\mu_i\}_{i=1}^n$ and $\sigma = \{\sigma_i\}_{i=1}^n$ are obtained by the feature encoder ϕ. However, the variational feature z is obtained through a reparameterization technique [56], where $z = \sigma * \epsilon + \mu$, $\epsilon = \{\epsilon_i\}_{i=1}^n$, and $\epsilon \sim N(0, I)$. Since the posterior distribution $p_\theta(z|s)$ is challenging to compute, variational inference is employed to model the posterior distribution by approximating it with another distribution $q_\phi(z|s) = N(\mu, \sigma^2)$ to minimize the Kullback–Leibler (KL) divergence, i.e., $\min D_{\mathrm{KL}}(q_\phi(z|s)||p_\theta(z|s))$. According to the definition of KL divergence, it can be expressed as follows:

$$\begin{aligned}
&\min D_{\mathrm{KL}}\left(q_\phi(z|s)||p_\theta(z|s)\right) \\
&\quad = \int q_\phi(z|s) \log \frac{q_\phi(z|s)}{p_\theta(z|s)}\, dz \\
&\quad = \log p_\theta(s) + \int q_\phi(z|s) \log q_\phi(z|s)\, dz \\
&\qquad - \int q_\phi(z|s) \log\left[p_\theta(s|z)p(z)\right] dz
\end{aligned} \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

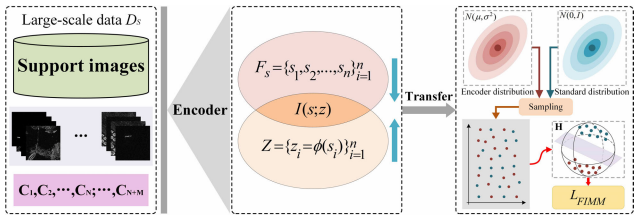


Fig. 8. Working principle of FIMM. First, support features F_S are extracted from support images, and an encoder is used to encode support samples into latent features Z with a Gaussian distribution. Next, an information bridge $I(s;z)$ with constraints is built between F_S and Z to maximize latent information. Finally, the objective function I is equivalently expressed as the measure in Hilbert space $\mathcal{H}$ between encoded latent samples and samples sampled from a standard Gaussian distribution, aiming to acquire maximum latent information.

where $p_\theta(s)$ is intractable, so minimizing $D_{\mathrm{KL}}(q_\phi(z|s)||p_\theta(z|s))$ is equivalent to minimizing the last two terms. Hence, we can transform this minimization term into the ELBO, comprising the log-likelihood (reconstruction) term $\mathcal{L}_{\mathrm{AE}}(s)$ and the regularization term $\mathcal{L}^{\mathrm{REG}}(s)$, which can be equivalently expressed as follows:

$$\begin{aligned}\mathcal{L}(\phi,\theta;s) &= \mathcal{L}_{\mathrm{AE}}(s)+\mathcal{L}^{\mathrm{REG}}(s)\\ &\equiv \mathbb{E}_{z\sim q_\phi(z|s)}\left[\log(p_\theta(s|z))\right]-D_{\mathrm{KL}}\left(q_\phi(z|s)||p(z)\right)\end{aligned} \tag{15}$$

where the purpose of maximizing $\mathcal{L}_{\mathrm{ELBO}}$ is to maximize the reconstruction loss $\mathcal{L}_{\mathrm{AE}}$ and minimize the KL divergence loss $\mathcal{L}^{\mathrm{REG}}$. By constraining the mean μ and variance σ, the generated latent distribution gradually approaches the standard Gaussian distribution, thereby achieving effective sampling in the latent space while mitigating fluctuations in class centers.

Despite stabilizing the positions of class centers, the dimensionality of F_S greatly exceeds that of Z. During modeling, the errors incurred in the F_S space are much larger than those in the Z space. Consequently, the model may tend to sacrifice the divergence on Z while minimizing the divergence on F_S, potentially hindering the efficient learning of the inference distribution $q_\phi(z|s)$. The underlying cause of this outcome lies in the low information transfer efficiency between F_S and Z. In addition, during feature aggregation, the amount of information conveyed by latent features is reduced.

To address the aforementioned issues and inspired by related studies on mutual information [51], we start from the mutual information $I(s;z)$ between input features F_S and latent variables Z. The proposed FIMM focuses more on high mutual information $I(s;z)$, encouraging the model to mitigate information bias. This facilitates effective information transmission and efficient utilization of feature information. The working principle is illustrated in Fig. 8.

First, we need to define the joint inference distribution $q_\phi(s,z)$ and the joint generative distribution $p_\theta(s,z)$ as follows:

$$q_\phi(s,z)\equiv p_D(s)q_\phi(z|s),\ p_\theta(s,z)\equiv p(z)p_\theta(s|z) \tag{16}$$

where $q_\phi(s,z)$ starts from the true data distribution $p_D(x)$ and learns the conditional distribution over z, while $p_\theta(s,z)$ starts from the prior distribution $p(z)$ and learns the conditional distribution over x. The corresponding conditional and marginal distributions can be defined as follows:

$$p_\theta(s)=\int_z p_\theta(s,z)\,dz,\ \ p_\theta(z|s)\propto p_\theta(s,z) \tag{17}$$

$$q_\phi(z)=\int_s q_\phi(s,z)\,ds,\ \ q_\phi(s|z)\propto q_\phi(s,z) \tag{18}$$

where $p_\theta(s)$ is the marginal of $p_\theta(s,z)$ with respect to x, and $q_\phi(z)$ is the marginal of $q_\phi(s,z)$ with respect to z. The ELBO can be equivalently expressed as follows:

$$\begin{aligned}\mathcal{L}(\phi,\theta;s)\equiv&-\mathbb{E}_{z\sim q_\phi(z|s)}\left[D_{\mathrm{KL}}\left(q_\phi(s|z)||p_\theta(s|z)\right)\right]\\ &-D_{\mathrm{KL}}\left(q_\phi(z)||p(z)\right).\end{aligned} \tag{19}$$

Next, we introduce a scaling parameter λ into the equivalent form $\mathcal{L}(\phi,\theta;s)$ to address the information imbalance between s and z. In addition, we incorporate another FIMM with an information preference parameter α to mitigate information preference issues [57]. The updated equivalent form can thus be defined as follows:

$$\begin{aligned}\mathcal{L}'(\phi,\theta;s)\equiv&-\mathbb{E}_{z\sim q_\phi(z|s)}\left[D_{\mathrm{KL}}\left(q_\phi(s|z)||p_\theta(s|z)\right)\right]\\ &-D_{\mathrm{KL}}\left(q_\phi(z)||p(z)\right)+\alpha I(s;z)\\ \equiv&\,\mathbb{E}_{p_D(s)}\mathbb{E}_{z\sim q_\phi(z|s)}\left[\log p_\theta(s|z)\right]\\ &-(1-\alpha)\mathbb{E}_{p_D(s)}D_{\mathrm{KL}}\left(q_\phi(z|s)||p(z)\right)\\ &-\left(\alpha+\lambda-1\right)D_{\mathrm{KL}}\left(q_\phi(z)||p(z)\right)\end{aligned} \tag{20}$$

where $I(s;z)$ is the mutual information between x and z under the distribution $q_\phi(s,z)$, λ is set to 100, and α is set to -0.5.

While the above objective function cannot be directly optimized, and $D_{\mathrm{KL}}(q_\phi(z)||p(z))$ is not easy to compute, it can be optimized using likelihood-free optimization techniques by first sampling $s\sim p_D(s)$ and then obtaining unbiased samples in $z\sim q_\phi(z|s)$. Therefore, $D_{\mathrm{KL}}(q_\phi(z)||p(z))$ can be replaced by any strict divergence $D(q_\phi(z)||p(z))$ to achieve further optimization. Minimizing $D(q_\phi(z)||p(z))$ alone allows us to maximize the mutual information between s and z. In this article, we adopt the concept of maximum mean discrepancy [58] to improve $D(q_\phi(z)||p(z))$. By mapping latent and prior samples to a high-dimensional Hilbert space $\mathcal{H}$ using a positive definite kernel function, we quantify the distance between their distributions by comparing all moments of the two sample distributions. This can be represented as follows:

$$\begin{aligned}&D^{\mathrm{FIMM}}\left[p,q,\mathcal{H}\right]\\ &:=\sup_{f\in\mathcal{H},||f||_{\mathcal{H}}\le 1}\left(\mathbb{E}_p\left[f(z)\right]-\mathbb{E}_q\left[f\left(\tilde{z}\right)\right]\right)\\ &=\mathbb{E}_{p(z),p(\tilde{z})}\left[k\left(z,\tilde{z}\right)\right]\\ &\quad+\mathbb{E}_{q(z),q(\tilde{z})}\left[k\left(z,\tilde{z}\right)\right]-2\mathbb{E}_{q(z),p(\tilde{z})}\left[k\left(z,\tilde{z}\right)\right]\end{aligned} \tag{21}$$

where latent support samples $Z=\{z_i=\phi(s_i)\}_{i=1}^n$ follow a Gaussian distribution $N(\mu,\sigma^2)$, and standard prior samples $\tilde{Z}=\{\tilde{z}_j\}_{j=1}^m$ follow a standard Gaussian distribution $N(0,I)$. As the distance between q and p decreases, it implies greater information between S and Z. In addition, $k(z,\tilde{z})=e^{-(||z-\tilde{z}||^2/2\sigma^2)}$ represents the Gaussian kernel function. Therefore, the third term of $\mathcal{L}'(\phi,\theta;s)$ can be simplified into the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

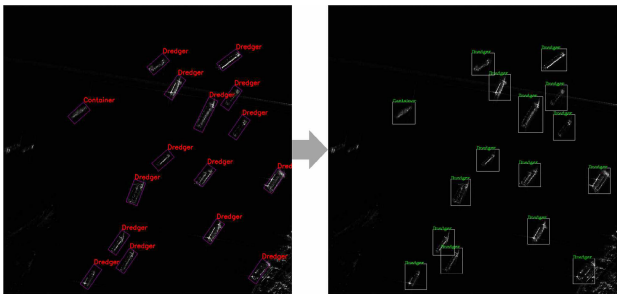


Fig. 9. Partial sample images from the SRSDD-v1.0 dataset.

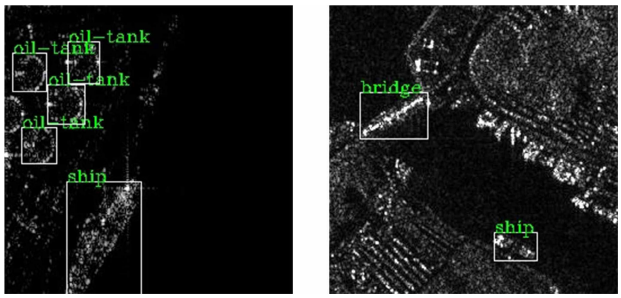


Fig. 10. Partial sample images from the MSAR-1.0 dataset.

constraint term loss of FIMM, denoted by $\mathcal{L}^{\mathrm{FIMM}}$, which can be expressed as follows:

$$\mathcal{L}^{\mathrm{FIMM}} = \frac{1}{n(n-1)} \sum_{i \neq j} k(z_i, z_j) - \frac{2}{nm} \sum_{i,j} k(z_i, \tilde{z}_j) + \frac{1}{m(m-1)} \sum_{i \neq j} k(\tilde{z}_i, \tilde{z}_j) \tag{22}$$

where a smaller value of $\mathcal{L}^{\mathrm{FIMM}}$ indicates a higher information transmission rate between S and Z, leading to a better feature aggregation effect.

E. Loss Function of IMFA

The total loss of IMFA can be formulated as follows:

$$\mathcal{L} = \mathcal{L}^{\mathrm{rpn}} + \mathcal{L}^{\mathrm{cls}} + \mathcal{L}^{\mathrm{reg}} + \mathcal{L}_{\mathrm{AE}} + \beta \cdot \mathcal{L}^{\mathrm{REG}} + \rho \cdot \mathcal{L}^{\mathrm{FIMM}} \tag{23}$$

where $\mathcal{L}^{\mathrm{rpn}}$ is the total loss of the region proposal network, $\mathcal{L}^{\mathrm{cls}}$ represents the classification loss, and $\mathcal{L}^{\mathrm{reg}}$ signifies the regression loss. $\mathcal{L}_{\mathrm{AE}}$ stands for reconstruction loss, $\mathcal{L}^{\mathrm{REG}}$ is the KL divergence loss, and $\mathcal{L}^{\mathrm{FIMM}}$ represents the mutual information constraint loss. The coefficients β and ρ are the weighting factors (default values: $\beta = 0.00025$ and $\rho = 0.1$).

V. Experiments and Analysis

In this section, we first introduce the SRSDD-v1.0 dataset and the MSAR dataset. Subsequently, we present the experimental setup and evaluation metrics. Finally, we analyze the relevant experiments of the proposed methods and innovative modules along with their own results.

A. Datasets

All experiments are conducted using SRSDD-v1.0 [59] and MSAR-1.0 [60] datasets to validate the proposed method's effectiveness. The pertinent parameters of the two used datasets are displayed in Table I in detail.

1) SRSDD-v1.0: The SRSDD-v1.0 dataset contains 666 images in total, sliced from 30 panoramic SAR images of port areas with a resolution of 1 m, which are all obtained from the GaoFen-3 satellite, and each image has 1024 × 1024 pixels. Among them, nearshore scenarios data occupy largely the dataset and are challenging to detect targets.

There are six different target types in the dataset, which are annotated with rotatable bounding boxes. To make it adapt to the few-shot SAR object detection task, we converted all the rotatable bounding boxes into the minimum circumscribed horizontal rectangular bounding boxes, as shown in Fig. 9. We split the dataset into the training, validating, and testing sets with a ratio of 8:1:1. Finally, we randomly separated this dataset into two base/novel splits according to the classes, with each split consisting of five base classes and 1 novel class, as illustrated in Table II.

2) MSAR-1.0: The MSAR-1.0 dataset originates from HISAE-1 and GaoFen-3 satellites and contains 28 449 SAR images with a resolution of 1 m. The dataset has an image size of 256 × 256 pixels, while some bridge images have 2048 × 2048 pixels. The dataset consists of four target types, with an enormous number of targets and severe imbalances in data among different target types. To make it adapt to the few-shot SAR target detection task, we chose 1000 SAR images, as shown in Fig. 10. Similarly, we split the obtained new dataset into training, validating, and testing sets with the same ratio of 8:1:1. Finally, we conducted the aforementioned base/novel splits operation, with each split containing three base classes and one novel class, as shown in Table II.

B. Implementation Details

Experiments in our proposed and comparative methods were both run on a personal computer with an AMD Ryzen 7 5800X 8-Core CPU, NVIDIA GeForce RTX 3070Ti GPU, and 16GB memory. Our proposed method was realized based on Mmdetection [61] using the PyTorch architecture. The ResNet101 [8] pretrained on the ImgaeNet was used as the backbone of our proposed method. Also, the optimizer was set to use the stochastic gradient descent (SGD) with a batch size of 2, a momentum of 0.9, and a weight decay of 0.0001.

During the basic training stage, the total iteration number was set to 110 000, and the learning rate was set initially to 0.0002, which dropped to 0.00002 and 0.000002 correspondingly at 73 000 iterations and 97 000 iterations. In the few-shot fine-tuning stage, the learning rate was set to 0.001, and performed fine-tuning under the conditions of K={1, 2, 3, 5, 10} shots at {400, 800, 1200, 1600, 2000} iterations.

C. Evaluation Metric

SRSDD-v1.0 and MSAR-1.0 are multiclass target detection datasets. In this study, we use average precision (AP) at an intersection over union (IoU) threshold of 0.5 and mean AP (mAP) as the primary evaluation metrics, along with macro-$F1$ and micro-$F1$ as additional metrics to assess the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED DESCRIPTION OF THE PARAMETERS OF THE TWO DIFFERENT DATASETS SRSDD-V1.0 AND MSAR-1.0

Dataset	Sensor	Polarization	Resolution	Classes	Number (Images)	Image Size (Pixel)
SRSDD-v1.0	Gaofen-3	HH, VV	1m	6	666	1024×1024
MSAR-1.0	HISEA-1, Gaofen-3	Full	1m	4	28449	256×256, 2048×2048

TABLE II
TWO DIFFERENT SPLIT SETTINGS FOR BASE/NOVEL ON SRSDD-V1.0 AND MSAR-1.0 DATASETS, RESPECTIVELY

Dataset	Split	Base class	Novel class
SRSDD-v1.0	1	Fishing, ore-oil, LawEnforce, Dredger, Container	Cell-Container
	2	Fishing, LawEnforce, Dredger, Container, Cell-Container	ore-oil
MSAR-1.0	1	bridge, airplane, oil-tank	ship
	2	airplane, oil-tank, ship	bridge

performance of multiclass target detection. The formulas for AP, mAP, macro-$F1$, and micro-$F1$ are as follows:

$$\mathrm{AP} = \int_0^1 P(R)d(R), \mathrm{mAP} = \frac{1}{n}\sum_{i=1}^{n} AP_i \tag{24}$$

$$P_i = \frac{TP_i}{TP_i + FP_i}, R_i = \frac{TP_i}{TP_i + FN_i} \tag{25}$$

$$P_{\mathrm{macro}} = \frac{\sum_{i=1}^{n} P_i}{n}, R_{\mathrm{macro}} = \frac{\sum_{i=1}^{n} R_i}{n} \tag{26}$$

$$P_{\mathrm{micro}} = \frac{\sum_{i=1}^{n} TP_i}{\sum_{i=1}^{n} TP_i + \sum_{i=1}^{n} FP_i}$$

$$R_{\mathrm{micro}} = \frac{\sum_{i=1}^{n} TP_i}{\sum_{i=1}^{n} TP_i + \sum_{i=1}^{n} FN_i} \tag{27}$$

$$F1_{\mathrm{macro}} = 2 \cdot \frac{P_{\mathrm{macro}} \cdot R_{\mathrm{macro}}}{P_{\mathrm{macro}} + R_{\mathrm{macro}}}$$

$$F1_{\mathrm{micro}} = 2 \cdot \frac{P_{\mathrm{micro}} \cdot R_{\mathrm{micro}}}{P_{\mathrm{micro}} + R_{\mathrm{micro}}} \tag{28}$$

where P_i represents the precision for the ith class, R_i represents the recall for the ith class, $P(R)$ denotes the precision obtained at a given recall rate R, n denotes the total number of target classes, i indexes the classes ranging from 1 to n, and AP_i represents the AP for the ith class. $F1$ represents the weighted average of P and R, where $F1_{\mathrm{macro}}$ is the average of the $F1$ scores for each class, and $F1_{\mathrm{micro}}$ aggregates the $F1$ scores across all classes. Higher values of AP, mAP, $F1_{\mathrm{macro}}$, and $F1_{\mathrm{micro}}$ indicate a better balance between precision and recall, signifying superior performance in few-shot target detection.

D. Comparison With State-of-the-Art Algorithms

To validate the superiority and effectiveness of IMFA, we conducted a performance comparison with state-of-the-art FSOD algorithms, including four metalearning methods [44], [45], [46], [48] and five transfer learning methods [22], [25], [27], [47], [62]. On the publicly available SRSDD-v1.0 dataset and MSAR-1.0 dataset, IMFA exhibits state-of-the-art detection performance.

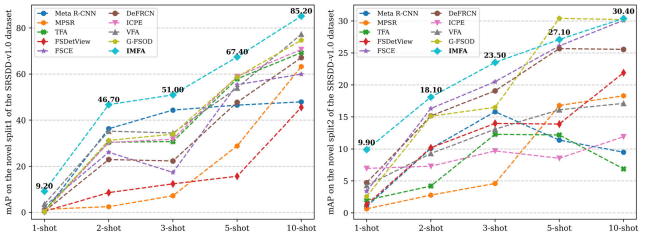


Fig. 11. In split 1 and split 2 settings of the SRSDD-v1.0 dataset, we visualize the performance comparison of the IMFA with some state-of-the-art few-shot target detection methods.

1) Results on SRSDD-v1.0 Dataset: As shown in Table III, we present the performance comparison of various FSOD methods on the SRSDD-v1.0 dataset, and IMFA outperforms significantly over other methods. Under both split 1 and split 2 settings, the novel class detection performance across different K-shot conditions achieved the best (9/10) and the second-best (1/10) results overall. For the overall performance across all K-shot conditions, IMFA's average detection performance for novel classes, base classes, and all classes significantly outperforms other algorithms. IMFA exhibits notable improvements, specifically with a 7.74% increase in novel class detection performance, a 3.4% increase in base class detection performance, and a 4.06% increase in all-class detection performance over the previous best results. This highlights IMFA's enhanced robustness and generalization capabilities in the few-shot scenario. In addition, the macro-$F1$ and micro-$F1$ scores are significantly higher than those of other methods, demonstrating that our approach is more competitive in multiclass SAR target detection tasks.

As illustrated in Fig. 11, for the SRSDD-v1.0 dataset, the left graph depicts the detection performance of novel classes under split 1 conditions, while the right graph shows the detection performance of novel classes under split 2 conditions. With an increase in the number of few-shot target samples, most methods exhibit an upward trend in the detection performance on novel classes. However, some methods show a declining trend in detection performance. Particularly, under scenarios with fewer or scarce samples, detection performance either improves slowly or deteriorates. This phenomenon is attributed to the complex background and interference from similar targets, making it challenging to extract the feature information of SAR targets. In addition, the instability of class centers further imposes constraints.

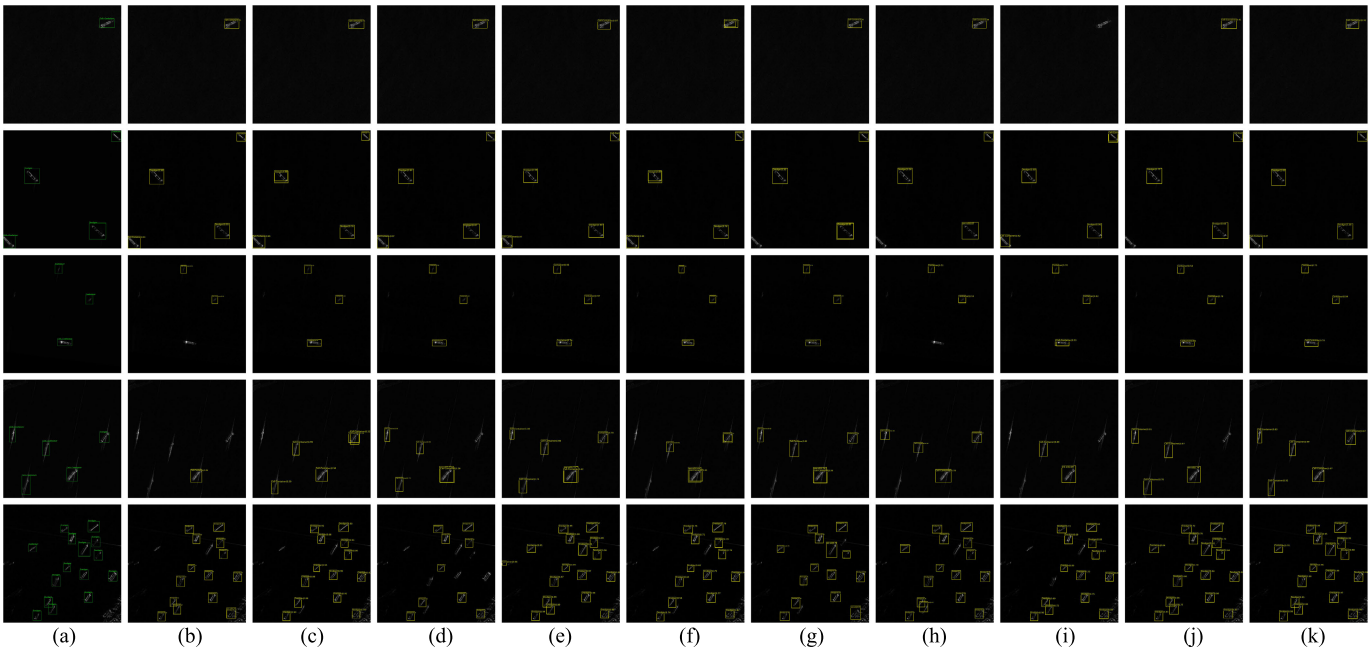
In contrast, our proposed IMFA adopts a more effective metalearning architecture. It leverages the context, multiscale information, and representative information of few-shot targets effectively through HMCA. Subsequently, FCM couples query-aware information to the support branch, enabling the model to better capture the feature information of novel class targets. Finally, under the Gaussian distribution conditions, our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE III

IN SPLIT 1 AND SPLIT 2 OF THE SRSDD-V1.0 DATASET, OUR IMFA WAS COMPARED WITH THE STATE-OF-THE-ART METHODS. RESULTS HIGHLIGHT THE BEST PERFORMANCE IN RED AND THE SECOND-BEST IN BLUE

Method	Backbone	Split 1 (Novel)					Split 2 (Novel)					*Novel*	*Base*	*All*	*Macro-F1*	*Micro-F1*
		1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*	*Avg.*	*Avg.*
Meta-Learning Methods																
FSDetView [44]	ResNet-101	0.44	8.60	12.39	15.68	45.58	1.30	10.20	14.00	13.90	21.90	14.07	48.09	42.47	30.07	33.05
Meta R-CNN [48]	ResNet-101	0.30	**36.35**	**44.34**	46.50	47.91	1.00	10.10	15.80	11.40	9.50	22.32	55.32	49.82	27.11	31.90
ICPE [45]	ResNet-101	0.23	30.37	31.81	**58.94**	70.67	**6.97**	7.31	9.70	8.55	11.93	23.65	53.44	48.47	20.32	22.15
VFA [46]	ResNet-101	**3.60**	35.30	34.50	53.98	**77.30**	4.30	9.30	13.10	16.10	17.10	26.46	60.06	54.46	37.31	40.84
Transfer-Learning Methods																
MPSR [62]	ResNet-101	1.30	2.48	7.23	28.86	63.26	0.63	2.79	4.60	16.77	18.31	14.62	37.22	33.03	20.06	19.75
TFA [47]	ResNet-101	0.90	30.50	30.80	57.90	69.30	2.00	4.20	12.30	12.20	6.90	22.70	59.88	53.69	34.76	37.23
DeFRCN [25]	ResNet-101	0.39	22.96	22.35	47.73	67.10	4.73	15.15	19.07	25.68	25.56	25.07	45.52	41.44	33.73	35.34
FSCE [22]	ResNet-101	2.60	26.13	17.39	55.40	59.95	3.40	**16.30**	**20.50**	26.10	30.10	25.79	54.60	49.80	33.21	36.98
G-FSOD [27]	ResNet-101	0.26	31.30	33.98	58.69	74.72	2.57	15.11	16.48	**30.41**	**30.23**	**29.38**	**60.50**	**55.32**	**37.84**	**42.07**
IMFA (Ours)	ResNet-101	**9.20**	**46.70**	**51.00**	**67.40**	**85.20**	**9.90**	**18.10**	**23.50**	**27.10**	**30.40**	**36.85**	**63.90**	**59.38**	**43.56**	**46.92**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 12. For the SRSDD-v1.0 dataset's split 1 setting, the detection performance of different methods under the ten-shot condition for base/novel class SAR targets. (a) Ground truth. (b) FSDetView. (c) Meta R-CNN. (d) ICPE. (e) VFA. (f) MSPR. (g) TFA. (h) DeFRCN. (i) FSCE. (j) G-FSOD. (k) IMFA (Ours).

constrained objective function FIMM maximizes latent information, leading to more robust feature aggregation. Especially in scenarios with extremely scarce novel class samples, IMFA can focus more on the inherent information of novel class targets, significantly improving their detection performance.

Furthermore, compared with state-of-the-art few-shot object detectors [Information-coupled prototype elaboration (ICPE), VFA, and generalized few-shot object detection (G-FSOD)], our proposed IMFA exhibits superior detection performance and stability.

In addition to the qualitative analysis mentioned above, we further conducted a visualization of the detection results for base/novel classes in the ten-shot setting under split 1 conditions, as shown in Fig. 12. From the detection results, it is evident that our proposed IMFA exhibits high detection performance in both simple and complex scenarios, accurately detecting base class and novel class targets. However, FSDetView and MPSR exhibited the poorest detection performance, with numerous instances of missed detections. In the visualizations, green labels represent ground truth target boxes, while yellow labels represent detected target boxes.

2) Results on MSAR-1.0 Dataset: As shown in Table IV, we also present the performance comparison of various FSOD methods on the MSAR-1.0 dataset. Specifically, whether based on metalearning or transfer learning, our IMFA demonstrates outstanding performance. Under both split 1 and split 2 settings, the detection performance on novel classes in different K-shot conditions generally achieves the best results (7/10) and the second-best results (2/10). In terms of the overall performance across all K-shot conditions, IMFA outperforms other algorithms in terms of average detection performance for novel classes, base classes, and all classes, with improvements

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

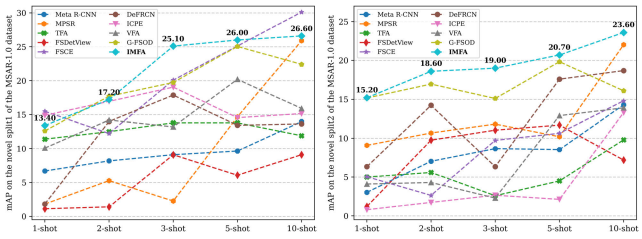


Fig. 13. In split 1 and split 2 settings of the MSAR-1.0 dataset, we visualize the performance comparison of the IMFA with some state-of-the-art few-shot target detection methods.

of 2.47%, 2.69%, and 4.63%, respectively, compared with the previous best results. Furthermore, compared with the state-of-the-art remote sensing few-shot object detector G-FSOD, our IMFA improves novel class detection performance by 2.47%, base class detection performance by 12.63%, and all-class detection performance by 10.1% under all K-shot settings. Furthermore, our macro-$F1$ and micro-$F1$ scores have also achieved a leading position. This reaffirms the superiority of our IMFA in remote sensing FSOD tasks.

As shown in Fig. 13, most algorithms exhibit oscillations in detection performance under few-shot conditions, while our IMFA steadily rises and demonstrates excellent performance. This is because IMFA can capture more intensive features of novel-class, few-shot SAR targets, and at the same time, enables the model to better adapt to novel-class targets.

We also conducted visualizations of the detection results for base and novel classes in the ten-shot setting under split 1 conditions, as depicted in Fig. 14. From the detection results, it is apparent that different object categories in this dataset vary significantly in scale. Nevertheless, IMFA can accurately detect targets of different scales simultaneously. However, FSDetView and meta R-CNN exhibited the poorest detection performance, with numerous instances of missed detections. In the visualizations, green labels represent ground truth target boxes, while yellow labels represent detected target boxes.

3) Comparison With Conventional Target Detection Algorithms: To validate the detection performance of the proposed method in the absence of few-shot settings, we conducted comparative experiments with advanced conventional object detection algorithms in the SAR domain (SOTA1 [63] and SOTA2 [64]), as well as with leading algorithms from the computer vision field (you only look once (YOLO) series [65], [66]). Under all data conditions, the ratio of training set, validation set, and test set was 8:1:1, exactly matching our proposed data setup. However, this setup completely ignores the few-shot setting. The results are shown in Table V. As can be seen from the table, our method performs exceptionally well in complex scenarios.

E. Ablation Studies

We conducted a comprehensive ablation study to analyze the actual impact of each innovative module in IMFA on detection performance, using both the SRSDD-v1.0 and MSAR-1.0 datasets. Specifically, we compared in detail by incrementally adding innovative modules to the baseline and provided an analysis of the overall performance changes. Detailed results are presented in Tables VI and VII.

1) Ablation Study of IMFA: We selected VFA as the baseline, which exhibits outstanding performance in natural images. However, in SAR scenarios, VFA's performance is mediocre, primarily due to the fundamental differences between SAR targets and objects in natural images. SAR images often contain complex background interference and targets with unclear information. The intricate contextual information significantly impacts the accurate representation of existing methods and confuses the feature information between targets and backgrounds. This results in the representative features of SAR targets being challenging to express efficiently. In addition, SAR targets exhibit high mobility, strong concealment, and rarity, resulting in highly unstable class centers for few-shot targets, making it difficult to effectively represent the latent features of SAR targets. This section will present a comprehensive performance comparison between the proposed IMFA and the baseline.

As shown in Table VI, we conducted the performance comparisons on the SRSDD-v1.0 dataset between the two data partitions, split 1 and split 2. In split 1 partition with $K = \{1, 2, 3, 5, 10\}$ shots, our IMFA demonstrated relative improvements of 5.6%, 11.4%, 16.5%, 13.42%, and 7.9%, respectively, over the baseline in the detection performance on novel classes, with an average improvement of 10.96%. In split 2 partition with $K = \{1, 2, 3, 5, 10\}$ shots, our IMFA exhibited relative improvements of 5.6%, 8.8%, 10.4%, 11%, and 13.3%, respectively, over the baseline in the detection performance on novel classes, with an average improvement of 9.82%. Looking at the comprehensive average detection levels of split 1 and split 2, there was an average improvement of 10.39% in the detection performance of novel classes, 3.84% in the detection performance of base classes, and 4.92% in the detection performance of all classes. The macro-$F1$ and micro-$F1$ scores have also increased by at least 6%.

As shown in Table VII, we conducted performance comparisons on the MSAR-1.0 dataset between the two data partitions, split 1 and split 2. In split 1 partition with $K = \{1, 2, 3, 5, 10\}$ shots, our IMFA demonstrated relative improvements of 2.6%, 3%, 11.9%, 5.8%, and 10.7%, respectively, over the baseline in the detection performance on novel classes, with an average improvement of 6.8%. In split 2 partition with $K = \{1, 2, 3, 5, 10\}$ shots, our IMFA exhibited relative improvements of 11.1%, 14.3%, 16.7%, 7.8%, and 9.7%, respectively, over the baseline in the detection performance on novel classes, with an average improvement of 11.92%. Considering the comprehensive average detection levels of split 1 and split 2, there was an average improvement of 9.36% in the detection performance of novel classes, 4.82% in the detection performance of base classes, and 5.97% in the detection performance of all classes. The macro-$F1$ and micro-$F1$ scores have also increased by at least 10%.

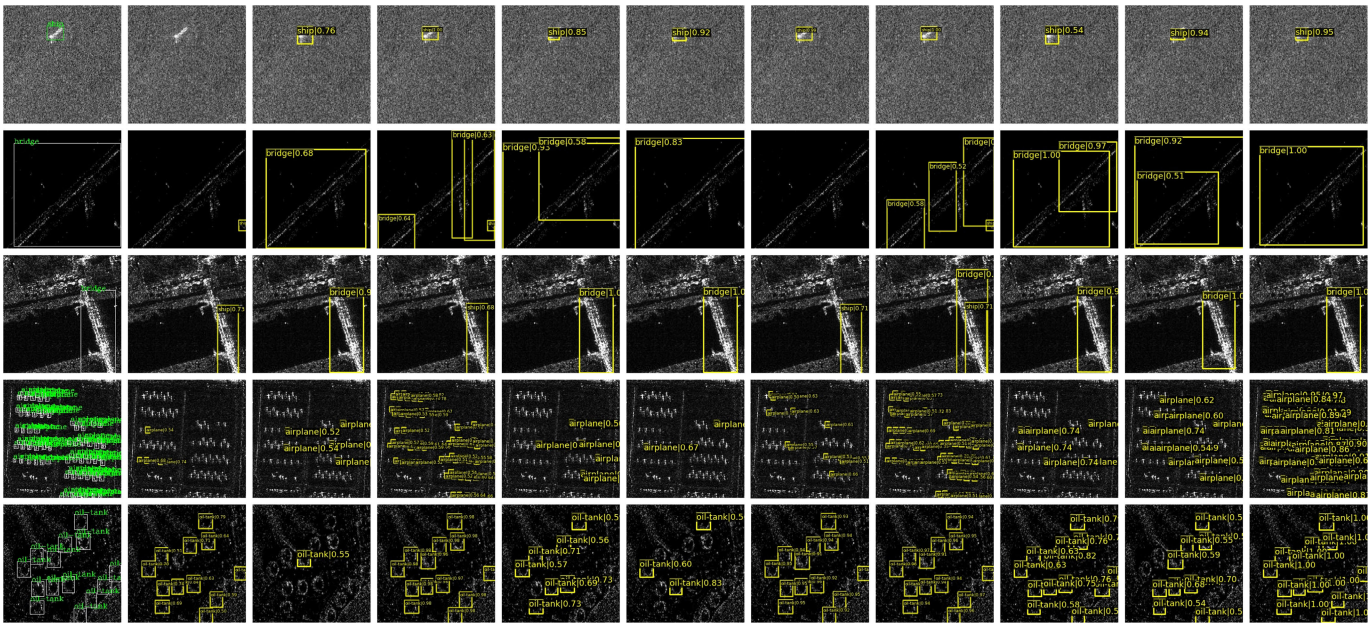
These results indicate that our proposed IMFA significantly enhances the detection performance of novel classes in few-shot SAR target scenarios, demonstrating the stability of class distribution and the ability to focus more on the representative information of SAR targets. This confirms the suitability of our IMFA for few-shot target detection tasks in SAR scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

IN SPLIT 1 AND SPLIT 2 OF THE MSAR-1.0 DATASET, OUR IMFA WAS COMPARED WITH STATE-OF-THE-ART METHODS. RESULTS HIGHLIGHT THE BEST PERFORMANCE IN RED AND THE SECOND-BEST IN BLUE

Method	Backbone	Split 1 (Novel)					Split 2 (Novel)					*Novel*	*Base*	*All*	*Macro-F1*	*Micro-F1*
		1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*	*Avg.*	*Avg.*
Meta-Learning Methods																
FSDetView [44]	ResNet-101	1.12	1.40	9.09	6.08	9.10	1.21	9.76	11.04	11.19	11.69	7.19	52.74	41.41	35.02	33.82
Meta R-CNN [48]	ResNet-101	6.70	8.19	9.10	9.64	13.99	3.03	7.04	8.66	8.54	14.29	8.92	63.72	49.44	37.04	35.52
ICPE [45]	ResNet-101	**14.88**	16.95	19.05	14.55	15.15	0.79	1.74	2.66	2.13	13.29	10.12	**68.43**	**53.85**	**50.37**	**43.87**
VFA [46]	ResNet-101	10.80	14.20	13.20	20.20	15.90	4.10	4.30	2.30	12.90	13.90	11.18	66.30	52.51	40.94	39.75
Transfer-Learning Methods																
TFA [47]	ResNet-101	11.40	12.50	13.80	13.80	11.90	5.00	5.60	2.60	4.50	9.80	9.09	61.16	48.14	39.12	39.01
MPSR [62]	ResNet-101	1.82	5.27	2.27	14.66	25.90	9.09	10.68	11.82	10.20	**22.03**	11.37	50.16	40.40	37.78	37.38
DeFRCN [25]	ResNet-101	1.82	13.99	17.88	13.43	13.64	6.34	14.22	6.43	17.59	18.69	12.40	50.50	40.97	40.81	40.83
FSCE [22]	ResNet-101	**15.48**	12.23	**20.08**	**25.11**	**30.12**	5.04	2.64	9.73	10.64	14.80	14.59	49.74	40.95	42.43	37.67
G-FSOD [27]	ResNet-101	12.61	**17.71**	19.73	25.05	22.40	**15.17**	**16.96**	**15.11**	**19.83**	16.10	**18.07**	58.49	48.38	43.88	42.93
IMFA (Ours)	ResNet-101	13.40	**17.20**	**25.10**	**26.00**	**26.60**	**15.20**	**18.60**	**19.00**	**20.70**	**23.60**	**20.54**	**71.12**	**58.48**	**52.61**	**50.71**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 14. For the MSAR-1.0 dataset's split 1 setting, the detection performance of different methods under the ten-shot condition for base/novel class SAR targets. (a) Ground truth. (b) FSDetView. (c) Meta R-CNN. (d) ICPE. (e) VFA. (f) MSPR. (g) TFA. (h) DeFRCN. (i) FSCE. (j) G-FSOD. (k) IMFA (Ours).

TABLE V

EXPERIMENTAL RESULTS UNDER DIFFERENT DATA SETTINGS

Method	Yolov5	Yolov6	Yolov7	Yolov8	SOTA1	SOTA2	**IMFA (Ours)**
SRSDD-v1.0	65.42	67.83	69.18	71.53	71.86	**72.68**	**73.82**
MSAR-1.0	73.68	76.96	78.73	**82.08**	80.16	81.62	**81.91**

2) Ablation Study of HMCA: The proposed HMCA aims to perform different tasks in different channel subspaces using a HMCA mechanism. It captures different data spatial relationships and collects contextual multiscale information in both horizontal and vertical directions, differentiating between targets and backgrounds. This addresses the issue of inaccurate expression of SAR target feature information in complex scenes under few-shot conditions. As evident from Tables VI and VII, significant improvements are observed, particularly in the 1-shot, 2-shot, and 3-shot scenarios, indicating notable enhancements in the detection performance of novel classes. Figs. 15 and 16 demonstrate that HMCA leads to stable performance improvement in detecting novel classes, mitigating the issue of performance fluctuations. This illustrates the method's ability to accurately extract representative features of novel SAR targets while avoiding interference from complex background information.

Furthermore, in split 1 setting of the SRSDD-v1.0 dataset, we conducted visual analysis of the changes in detection performance mAP and recall for both base and novel classes under the conditions of 3-shot, 5-shot, and 10-shot, as shown in Figs. 17 and 18. Figs. 17(b) and 18(b) reveal that compared with the baseline, our base class detection performance and corresponding recall show slight improvements. However,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF IMFA UNDER SPLIT 1 AND SPLIT 2 SETTINGS OF THE SRSDD-V1.0 DATASET

Method	HMCA	FCM	FIMM	Split 1 (Novel)					Split 2 (Novel)					*Novel*	*Base*	*All*	*Macro-F1*	*Micro-F1*
				1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*	*Avg.*	*Avg.*
VFA (baseline)				3.60	35.30	34.50	53.98	77.30	4.30	9.30	13.10	16.10	17.10	26.46	60.06	54.46	37.31	40.84
IMFA (Ours)	✓			4.30	40.20	41.30	60.90	79.00	5.30	14.30	18.10	20.50	21.50	30.54	59.06	54.31	38.16	41.52
	✓	✓		7.10	45.20	43.10	65.30	82.70	6.80	15.20	22.60	26.20	27.10	34.13	57.43	53.55	41.29	44.67
	✓	✓	✓	**9.20**	**46.70**	**51.00**	**67.40**	**85.20**	**9.90**	**18.10**	**23.50**	**27.10**	**30.40**	**36.85**	**63.90**	**59.38**	**43.56**	**46.92**
Improvement				**+5.60**	**+11.40**	**+16.50**	**+13.42**	**+7.90**	**+5.6**	**+8.8**	**+10.40**	**+11.00**	**+13.30**	**+10.39**	**+3.84**	**+4.92**	**+6.25**	**+6.08**

TABLE VII

ABLATION STUDIES WERE CONDUCTED ON EACH MODULE OF IMFA UNDER SPLIT 1 AND SPLIT 2 SETTINGS OF THE MSAR-1.0 DATASET

Method	HMCA	FCM	FIMM	Split 1 (Novel)					Split 2 (Novel)					*Novel*	*Base*	*All*	*Macro-F1*	*Micro-F1*
				1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot	*Avg.*	*Avg.*	*Avg.*	*Avg.*	*Avg.*
VFA (baseline)				10.80	14.20	13.20	20.20	15.90	4.10	4.30	2.30	12.90	13.90	11.18	66.30	52.51	40.94	39.75
IMFA (Ours)	✓			11.40	14.70	18.20	19.60	20.70	5.90	10.80	12.10	14.90	15.70	14.40	69.85	56.00	46.72	44.69
	✓	✓		12.60	14.90	21.00	22.90	24.40	9.20	14.90	15.20	16.40	19.60	17.11	70.56	57.18	49.13	48.89
	✓	✓	✓	**13.40**	**17.20**	**25.10**	**26.00**	**26.60**	**15.20**	**18.60**	**19.00**	**20.70**	**23.60**	**20.54**	**71.12**	**58.48**	**52.61**	**50.71**
Improvement				**+2.60**	**+3.00**	**+11.90**	**+5.80**	**+10.70**	**+11.10**	**+14.30**	**+16.70**	**+7.80**	**+9.70**	**+9.36**	**+4.82**	**+5.97**	**+11.67**	**+10.96**

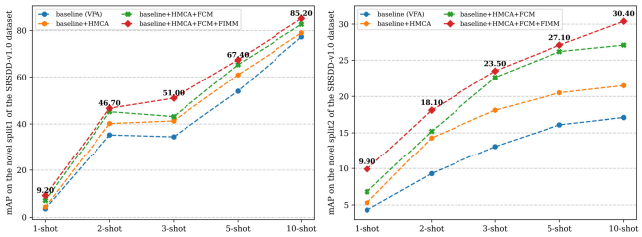


Fig. 15. In split 1 and split 2 settings of the SRSDD-v1.0 dataset, we visualize the performance comparison between IMFA and the baseline method.

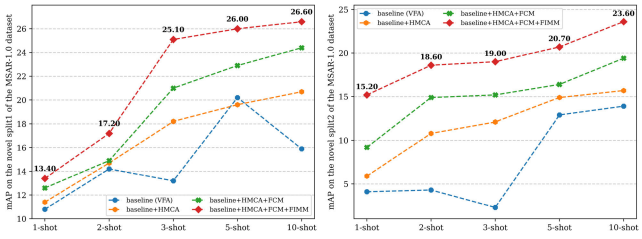


Fig. 16. In split 1 and split 2 settings of the MSAR-1.0 dataset, we visualize the performance comparison between IMFA and the baseline method.

there is a substantial improvement in both the detection performance and recall of novel classes, especially in the three-shot and five-shot conditions, where the mAP sees an average improvement of 6.86%. This demonstrates that the proposed HMCA maintains excellent performance even under conditions of extremely limited samples, affirming its efficacy in more effectively extracting representative features from few-shot SAR targets.

3) Ablation Study of FCM: The proposed FCM aims to enhance the model's perceptual ability for query images, reducing information interference among different SAR targets. When encountering novel SAR target classes, FCM enables the rapid adaptation of base-class knowledge to novel class targets. This addresses issues such as the model's low sensitivity to representative features in support images and insufficient awareness of query images under few-shot conditions. As evident from Tables VI and VII, there is a substantial improvement in the detection performance of novel classes. Figs. 15 and 16 reveal that, under the 3-shot, 5-shot, and 10-shot conditions, FCM leads to even greater improvements in the detection performance of novel classes. This demonstrates the method's effectiveness in reducing information interference among different targets within a limited number of SAR samples.

Furthermore, in split 1 setting of the SRSDD-v1.0 dataset, we conducted visual analysis of changes in detection performance mAP and recall for both base and novel classes under the conditions of 3-shot, 5-shot, and 10-shot, as shown in Figs. 17 and 18. Figs. 17(c) and 18(c) indicate that, compared with the baseline with the introduction of HMCA, our novel classes' detection performance and recall exhibit stable improvements, especially with a 4.4% increase in mAP under the five-shot condition. This underscores that the proposed FCM can further enhance the detection performance of few-shot SAR targets, effectively distinguishing different SAR targets and endowing the model with superior generalization capabilities.

4) Ablation Study of FIMM: The proposed FIMM aims to establish an information bridge between support features and latent features under the premise of constructing a Gaussian distribution. This aims to address issues such as large fluctuations in class centers and information imbalances for SAR samples under extremely limited conditions, thus maximizing the utilization of latent features. As evident from Table VI and Table VII, there is a significant improvement in the detection performance of novel classes. Figs. 15 and 16 indicate that the performance improvement is relatively stable in the SRSDD-v1.0 dataset, but more pronounced in the MSAR-1.0 dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

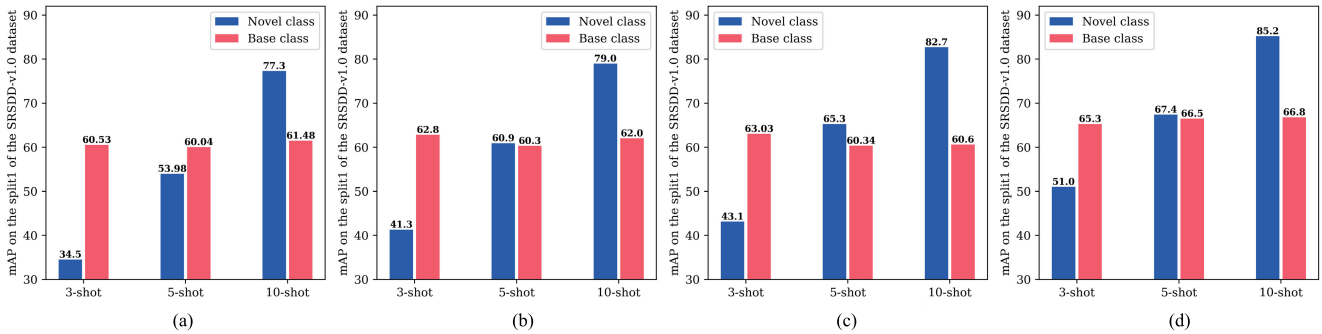


Fig. 17. In split 1 setting of the SRSDD-v1.0 dataset, we visualize the mAP of base and novel classes under the 3-shot, 5-shot, and 10-shot conditions. (a) Baseline. (b) Baseline + HMCA. (c) Baseline + HMCA + FCM. (d) Baseline + HMCA + FCM + FIMM.

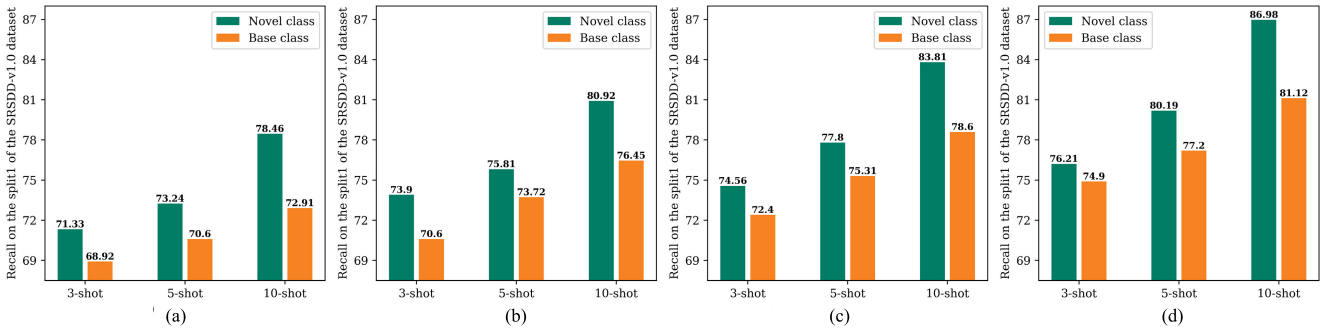


Fig. 18. In split 1 setting of the SRSDD-v1.0 dataset, we visualize the recall of base and novel classes under the 3-shot, 5-shot, and 10-shot conditions. (a) Baseline. (b) Baseline + HMCA. (c) Baseline + HMCA + FCM. (d) Baseline + HMCA + FCM + FIMM.

This demonstrates the method's ability to efficiently express SAR target features in complex scenes with scarce samples.

Furthermore, in split 1 setting of the SRSDD-v1.0 dataset, we conducted visual analysis of changes in detection performance mAP and recall for both base and novel classes under the conditions of 3-shot, 5-shot, and 10-shot, as shown in Figs. 17 and 18. Figs. 17(d) and 18(d) show that, compared with the previous stacked innovative modules, both the detection performance and recall of base and novel classes have been significantly improved, particularly with a 7.9% increase in mAP under the three-shot condition. This underscores that under extremely limited sample conditions, the proposed FIMM can stabilize class centers more effectively and efficiently represent the latent features of SAR targets, thereby mitigating information shifts.

5) Hyperparameter Analysis: We analyzed the hyperparameters in the IMFA loss function to evaluate their impact on the proposed method. Since β was selected as the optimal choice after multiple experiments in the baseline paper, we continue to set β at 0.00025. ρ is a hyperparameter of the FIMM loss, which will affect the representation of latent feature information. We made several empirical judgments about hyperparameter ρ and selected its values within an appropriate range. The detection performance is shown in Table VIII. As the results indicate, when ρ is around 0.1, the detection performance is optimal in most cases, whereas when ρ is around 0.01, the performance tends to be second best.

6) Computational Efficiency: To evaluate the efficiency of our proposed method in processing SAR images, we performed a quantitative analysis comparing model parameter size, training time, and inference time with the baseline method. The complexity of training time is the duration per iteration, including both the base class and novel class training complexities. The inference time complexity is the time it takes for the model to detect on a 1024 SAR image, which includes both the base class and novel class inference complexities. The results, shown in Tables IX and X, indicate that our proposed IMFA method does not significantly increase the computational complexity of the model. Compared with the baseline method, the total parameter size of the model has only slightly increased, and the complexities of training and inference times for both phases show only minor fluctuations, which could be affected by factors such as memory usage during model initialization.

TABLE VIII

DETECTION ACCURACY FOR NOVEL CLASSES UNDER DIFFERENT HYPERPARAMETER SETTINGS WITH A TEN-SHOT CONDITION

Hyperparameter		ρ				
		10	1	0.1	0.01	0.001
SRSDD-v1.0	β=0.00025	80.01	82.72	**85.20**	**83.22**	81.80
MSAR-1.0	β=0.00025	24.50	25.52	**26.60**	**25.90**	24.85

TABLE IX

MODEL PARAMETER SIZE AND BASE CLASS TRAINING/ INFERENCE TIME COMPLEXITIES

Method	Params	Training (s/iter)	Inference (s/iter)
Baseline	75.82M	0.2254	0.2686
IMFA (Ours)	80.94M	0.2317	0.2686

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

TABLE X
NOVEL CLASS TRAINING/INFERENCE TIME COMPLEXITIES

Method	Training (s/iter)					Inference (s/iter)				
	1-shot	2-shot	3-shot	5-shot	10-shot	1-shot	2-shot	3-shot	5-shot	10-shot
Baseline	0.2054	0.2055	0.2053	0.2053	0.2054	0.2685	0.2686	0.2687	0.2686	0.2687
IMFA(Ours)	0.2104	0.2105	0.2103	0.2106	0.2107	0.2685	0.2687	0.2687	0.2686	0.2686

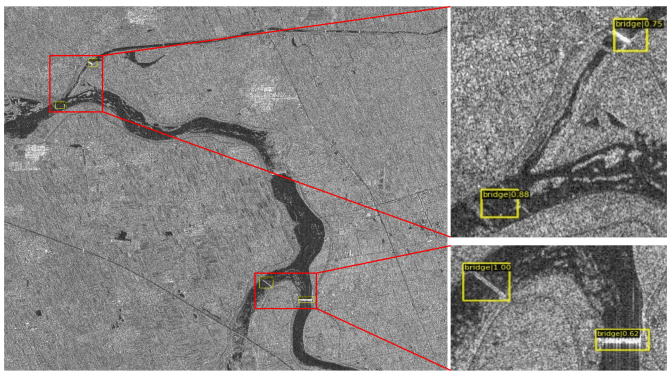


Fig. 19. Detection results for large-scale SAR images.

In addition, we also evaluated the computational efficiency of our proposed method on large-scale SAR images. Within split 2 setting of the MSAR-1.0 dataset, we selected additional test data comprising large-scale SAR images of size 2048 × 2048. Under ten-shot conditions, our method can accurately detect few-shot SAR targets in these large-scale images, with the inference time for one large-scale SAR image being approximately 0.88 s. The detection results are illustrated in Fig. 19.

VI. CONCLUSION

This article addresses the few-shot target detection task in SAR scenes and proposes a method for few-shot target detection in SAR imagery via IMFA. Our proposed method efficiently captures representative features of SAR targets and utilizes it effectively. We design an HMCA to obtain global context multiscale information and representative features of different targets for distinguishing targets from the background. In addition, we propose a FCM to enhance the model's cognitive ability for query images to improve its generalization capacity. Furthermore, we address the problem of drastic fluctuations in class centers by modeling SAR samples with Gaussian distributions. Building upon this, we introduce FIMM as a constraint function to address the information imbalance between support features and latent features, thereby maximizing the utilization of latent features comprehensively. Experiments on the SRSDD-v1.0 and MSAR-1.0 datasets demonstrate that our proposed method outperforms the state-of-the-art few-shot target detection methods in terms of detection performance.

Due to the potential significant differences between novel-class few-shot SAR targets and base-class SAR targets, novel-class SAR targets may be difficult to detect, and base-class knowledge may not generalize well to novel-class targets. This can lead to catastrophic forgetting of base-class knowledge and poor detection performance for novel-class targets. Therefore, in future work, we will explore representation learning methods for novel and base classes under different distribution conditions to avoid catastrophic forgetting of base-class knowledge, enhance the generalization capability of base-class knowledge, and improve the overall detection performance.

ACKNOWLEDGMENT

The authors would like to thank the editors and reviewers for their the insightful and helpful comments.

REFERENCES

[1] K.-S. Chen, *Principles of Synthetic Aperture Radar Imaging: A System Simulation Approach*, vol. 2. Boca Raton, FL, USA: CRC Press, 2016.
[2] B. Li, B. Liu, L. Huang, W. Guo, Z. Zhang, and W. Yu, "OpenSARShip 2.0: A large-volume dataset for deeper interpretation of ship targets in Sentinel-1 imagery," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–5.
[3] A. Kaur and S. Singh Kang, "Sea ice detection using synthetic aperture radar algorithm in image processing," *Int. J. Comput. Appl.*, vol. 121, no. 13, pp. 1–5, Jul. 2015.
[4] H. Sun, M. Shimada, and F. Xu, "Recent advances in synthetic aperture radar remote sensing—Systems, data processing, and applications," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2013–2016, Nov. 2017.
[5] G. Gao, L. Liu, L. Zhao, G. Shi, and G. Kuang, "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 6, pp. 1685–1697, Jun. 2009.
[6] X. Leng, K. Ji, K. Yang, and H. Zou, "A bilateral CFAR algorithm for ship detection in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1536–1540, Jul. 2015.
[7] M.-D. Li, X.-C. Cui, and S.-W. Chen, "Adaptive superpixel-level CFAR detector for SAR inshore dense ship detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[8] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[10] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
[11] Z. Su et al., "Lightweight pixel difference networks for efficient visual representation learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14956–14974, Dec. 2023.
[12] Z.-Q. Zhao, P. Zheng, S.-T. Xu, and X. Wu, "Object detection with deep learning: A review," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 11, pp. 3212–3232, Nov. 2019.
[13] L. Liu et al., "Deep learning for generic object detection: A survey," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 261–318, Feb. 2020.
[14] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

[15] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet++ for object detection," in *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3509–3521, 2024, doi: 10.1109/TPAMI.2023.3342120.

[16] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.

[17] Z. Huang, Z. Pan, and B. Lei, "What, where, and how to transfer in SAR target recognition based on deep CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2324–2336, Apr. 2020.

[18] T. Zhang et al., "Balance learning for ship detection from synthetic aperture radar remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 190–207, Dec. 2021.

[19] Z. Zhou, Z. Cui, Z. Zang, X. Meng, Z. Cao, and J. Yang, "UltraHi-PrNet: An ultra-high precision deep learning network for dense multi-scale target detection in SAR images," *Remote Sens.*, vol. 14, no. 21, p. 5596, Nov. 2022.

[20] J. Zhou et al., "DiffDet4SAR: Diffusion-based aircraft target detection network for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[21] S. Sun, S. Zhi, Q. Liao, J. Heikkilä, and L. Liu, "Unbiased scene graph generation via two-stage causal modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12562–12580, Mar. 2023.

[22] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[23] Q. Zeng and J. Geng, "Task-specific contrastive learning for few-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 143–154, Sep. 2022.

[24] Q. Fan, W. Zhuo, C.-K. Tang, and Y.-W. Tai, "Few-shot object detection with attention-RPN and multi-relation detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4013–4022.

[25] L. Qiao, Y. Zhao, Z. Li, X. Qiu, J. Wu, and C. Zhang, "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8681–8690.

[26] B. Li, C. Wang, P. Reddy, S. Kim, and S. Scherer, "AirDet: Few-shot detection without fine-tuning for autonomous exploration," in *Proc. 17th Eur. Conf. Computer Vis.*, Oct. 2022, pp. 427–444.

[27] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[28] Z. Yue, F. Gao, Q. Xiong, J. Sun, A. Hussain, and H. Zhou, "A novel few-shot learning method for synthetic aperture radar image recognition," *Neurocomputing*, vol. 465, pp. 215–227, Nov. 2021.

[29] L. Zhang et al., "Domain knowledge powered two-stream deep network for few-shot SAR vehicle recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5215315.

[30] X. Sun, Y. Lv, Z. Wang, and K. Fu, "SCAN: Scattering characteristics analysis network for few-shot aircraft classification in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226517.

[31] Y. Tai, Y. Tan, S. Xiong, Z. Sun, and J. Tian, "Few-shot transfer learning for SAR image classification without extra SAR samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2240–2253, 2022.

[32] M. Yang, X. Bai, L. Wang, and F. Zhou, "HENC: Hierarchical embedding network with center calibration for few-shot fine-grained SAR target classification," *IEEE Trans. Image Process.*, vol. 32, pp. 3324–3337, 2023.

[33] H. Ren, X. Yu, S. Liu, L. Zou, X. Wang, and H. Tang, "Adaptive convolutional subspace reasoning network for few-shot SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, doi: 10.1109/TGRS.2023.3288987.

[34] X. Wang, H. Zhou, Z. Chen, J. Bai, J. Ren, and J. Shi, "Few-shot SAR ship image detection using two-stage cross-domain transfer learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2195–2198.

[35] S. Chen, J. Zhang, R. Zhan, R. Zhu, and W. Wang, "Few shot object detection for SAR images via feature enhancement and dynamic relationship modeling," *Remote Sens.*, vol. 14, no. 15, p. 3669, Jul. 2022.

[36] Z. Zhou et al., "FSODS: A lightweight metalearning method for few-shot object detection on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, doi: 10.1109/TGRS.2022.3192996.

[37] Z. Deng, H. Sun, S. Zhou, and J. Zhao, "Learning deep ship detector in SAR images from scratch," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4021–4039, Jun. 2019.

[38] J. Cui, H. Jia, H. Wang, and F. Xu, "A fast threshold neural network for ship detection in large-scene SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6016–6032, 2022.

[39] Y. Zhou, X. Jiang, G. Xu, X. Yang, X. Liu, and Z. Li, "PVT-SAR: An arbitrarily oriented SAR ship detector with pyramid vision transformer," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 291–305, 2023.

[40] K. Li, M. Zhang, M. Xu, R. Tang, L. Wang, and H. Wang, "Ship detection in SAR images based on feature enhancement swin transformer and adjacent feature fusion," *Remote Sens.*, vol. 14, no. 13, p. 3186, Jul. 2022.

[41] G. Han, S. Huang, J. Ma, Y. He, and S.-F. Chang, "Meta faster R-CNN: Towards accurate few-shot object detection with attentive feature alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 780–789.

[42] Z. Yang, C. Zhang, R. Li, Y. Xu, and G. Lin, "Efficient few-shot object detection via knowledge inheritance," *IEEE Trans. Image Process.*, vol. 32, pp. 321–334, 2023.

[43] G. Zhang, Z. Luo, K. Cui, S. Lu, and E. P. Xing, "Meta-DETR: Image-level few-shot detection with inter-class correlation exploitation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12832–12843, 2023, doi: 10.1109/TPAMI.2022.3195735.

[44] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2023.

[45] X. Lu et al., "Breaking immutable: Information-coupled prototype elaboration for few-shot object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2023, vol. 37, no. 2, pp. 1844–1852.

[46] J. Han, Y. Ren, J. Ding, K. Yan, and G.-S. Xia, "Few-shot object detection via variational feature aggregation," 2023, *arXiv:2301.13411*.

[47] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[48] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9577–9586.

[49] R. Amit and R. Meir, "Meta-learning by adjusting priors based on extended PAC-Bayes theory," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 205–214.

[50] J. Xu, H. Le, M. Huang, S. Athar, and D. Samaras, "Variational feature disentangling for fine-grained few-shot classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8812–8821.

[51] B. Poole, S. Ozair, A. Van Den Oord, A. Alemi, and G. Tucker, "On variational bounds of mutual information," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 5171–5180.

[52] W. Su et al., "Towards all-in-one pre-training via maximizing multi-modal mutual information," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 15888–15899.

[53] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA. Red Hook, NY, USA: Curran Associates Inc., 2017, pp. 6000–6010.

[54] H. Cheng, H. Wu, J. Zheng, K. Qi, and W. Liu, "A hierarchical self-attention augmented Laplacian pyramid expanding network for change detection in high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 52–66, Dec. 2021.

[55] H. Hu, S. Bai, A. Li, J. Cui, and L. Wang, "Dense relation distillation with context-aware aggregation for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10185–10194.

[56] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[57] I. Higgins et al., "beta-vae: Learning basic visual concepts with a constrained variational framework," in *Proc. ICLR*, vol. 3, 2017.

[58] Y. Li, K. Swersky, and R. Zemel, "Generative moment matching networks," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1718–1727.

[59] S. Lei, D. Lu, X. Qiu, and C. Ding, "SRSDD-v1.0: A high-resolution SAR rotation ship detection dataset," *Remote Sens.*, vol. 13, no. 24, p. 5104, Dec. 2021.

[60] R. Xia et al., "CRTransSar: A visual transformer based on contextual joint representation learning for SAR ship detection," *Remote Sens.*, vol. 14, no. 6, p. 1488, Mar. 2022.

[61] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.

[62] J. Wu, S. Liu, D. Huang, and Y. Wang, "Multi-scale positive sample refinement for few-shot object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Oct. 2020, pp. 456–472.

[63] W. Du et al., "SARNas: A hardware-aware SAR target detection algorithm via multiobjective neural architecture search," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–23, 2023, doi: 10.1109/TGRS.2023.3292618.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.

[64] Y. Liu, G. Yan, F. Ma, Y. Zhou, and F. Zhang, "SAR ship detection based on explainable evidence learning under intraclass imbalance," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024, doi: 10.1109/TGRS.2024.3373668.
[65] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.
[66] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.



Zheng Zhou (Graduate Student Member, IEEE) received the B.E. degree from Chengdu University of Technology, Chengdu, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu.

His research interests include synthetic aperture radar (SAR) image processing, target detection, and few-shot learning.



Zongjie Cao (Member, IEEE) received the B.E. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1999 and 2005, respectively.

From 2006 to 2008, he was a Post-Doctoral Researcher with the Communication and Information System Postdoctoral Center, University of Electronic Science and Technology of China (UESTC), Chengdu, China. In 2008, he joined the School of Electronic Engineering, UESTC, where he is currently a Professor with the School of Information and Communication Engineering. He has authored or coauthored over 100 articles. His research interests include radar signal processing, synthetic aperture radar imaging, and target recognition.

Dr. Cao is a reviewer for several international journals and conferences, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and the *IET Radar, Sonar & Navigation, Remote Sensing*.



Qin Chen (Graduate Student Member, IEEE) received the B.E. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2020, where he is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering.

His research interests include the design of radar system, radar signal processing, gesture recognition, and deep learning.



Kailing Tang (Graduate Student Member, IEEE) received the B.E. degree in information engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2022, where he is currently pursuing the M.S. degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar (SAR) image processing and object recognition.



Yujian Li (Graduate Student Member, IEEE) received the B.E. degree in electronic information engineering from the Glasgow College, University of Electronic Science and Technology of China, Chengdu, China, in 2023, where he is currently pursuing the M.S. degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar (SAR) image processing and open-set target detection.



Yiming Pi (Senior Member, IEEE) received the master's degree in communication and information system and the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China, Chengdu, China, in 1990 and 1993, respectively.

He is currently a Professor with the School of Electronic Engineering, University of Electronic Science and Technology of China. In 1994, he began his post-doctoral research in electronic and communication at Beijing University of Posts and Telecommunications, Beijing, China. In 1997, he returned to the University of Electronic Science and Technology of China, where he is currently with the College of Electronic Engineering. His research interests include terahertz technology, synthetic aperture radar, and radar signal processing.



Zongyong Cui (Member, IEEE) received the B.E. and Ph.D. degrees in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2007 and 2015, respectively.

From 2013 to 2014, he was a Visiting Student with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar image processing, target recognition, and machine learning.

Dr. Cui is a reviewer for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSATR), and *IET Radar, Sonar & Navigation*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:56 UTC from IEEE Xplore. Restrictions apply.